Zebra CV70

Product Reference Guide

Zebra Technologies | 3 Overlook Point | Lincolnshire, IL 60069 USA
zebra.com

MN-005426-0EN Rev A

Copyright

This document contains proprietary information and is intended for internal use only.

Contents

1. **Release Notes**
2. **Release Notes: Zebra Aurora Focus 10.0.15**
3. **Product Description**
4. **Description**
5. **Description**
6. **Description**
7. **Description**
8. **Description**
9. **Description**
10. **Description**
11. **Description**
12. **Description**
13. **Description**
14. **Description**
15. **Description**
16. **Description**
17. **Description**
18. **Description**

Release Notes

Use this section to track feature updates, bug fixes, and other changes by release.

How to add a new release note

1. Copy `docs/release-notes/_release-notes-template.mdx` to a new `.md` or `.mdx` file in this folder.
2. Name it with a sortable release identifier where newer releases sort higher (example: `2026-03-01-aurora-focus-10-0-123.mdx`).
3. Replace all placeholder values (`[Product Name]`, `[Version]`, etc.) and publish.

Release notes are shown in descending file name order, so the most recent release appears at the top.

Download PDF

Release Notes: Zebra Aurora Focus 10.0.15

This release of Zebra Aurora Focus (V10.0.15) introduces significant enhancements, including the ability to save intermediate images, view timestamps in the results log, decode multiple 1D barcodes simultaneously, and adds support for the Zebra Integrated Multi-Function Light (ZIML) on FS42 2MP devices. It also features improvements to JavaScript capabilities and overall system stability.

Important Notes

- **FS80 Upgrade/Downgrade:** Full upgrades contain large files that require longer installation times. Do not pull power to the device during the upgrade. After upgrading the FS80 to the 10.0 release, downgrades can no longer be completed with an incremental update package and must be done with the full upgrade package. To downgrade from 10.0, install the 9.1 Full Update Firmware first, and then to the required firmware if different than 9.1.
- **Rolling Shutter Support:** Rolling Shutter is no longer supported on xS40-70 (5MP) devices.

Software and Firmware Versions

This table provides a clear reference for all component versions included in the release bundle.

Component	Version
Aurora Focus	10.0.15
Aurora Focus HMI	10.0.33
Connectivity Gateway	1.3.6
Aurora Deep Learning	9.4.11193
(BSP) GS20	10.0.723 - CAAGDS00-008-R1
(BSP) FS40, VS40, FS70, VS70	10.0.723 - CAAESS00-008-R1
(BSP) FS42, NS42	10.0.723 - CAAGGS00-008-R1
(BSP) FS10, FS20, VS20	10.0.723 - CAAFFS00-0068R1
(BSP) FS80	10.0.414 - SAAHMS00-002-R01

New Features

- **Save Image Enhancements:** `[DCH-14673]` Save intermediate images for continuous and aggregate modes that process multiple frames during execution. This is useful for troubleshooting partial image sequences to understand why misdecodes occur.
- **View Timestamps in the Results Log:** `[DCH-20859]` Timestamps are now viewable to the millisecond in the Results Log, aligning with the results history in the WebHMI.
- **Decode Multiple 1D Barcodes Simultaneously:** `[DCH-25695]` Datacode now supports simultaneous decode of multiple 1D symbologies. The Datacode tool is also now compatible with ImagePerfect, providing the ability to program a sequence of read attempts with different acquisition banks.
- **Byte Swap Support:** `[DCH-23899]` Support to access the Industrial Ethernet data structure to toggle the byte order before sending it, supporting Keyence PLC. The byte swap option is provided for all string type IE attributes.
- **Zebra Integrated Multi-Function Light (ZIML) Support on FS42 2MP:** `[DCH-26089]` Support for using FS42 2MP devices with the ZIML without Hidden Strobe.
- **Sensor Windowing:** `[DCH-12982]` Image sensor partial scanning and sensor ROI windowing improves speed for high-speed applications by reducing image acquisition time and enhancing image transfer efficiency.
- **Callback Functions Using JavaScript:** `[DCH-24103]` Added JavaScript callback support for GPIO, TCPIP and Serial port imports.
- **Additional JavaScript Enhancements:**
  - Binarization, Brightness, Pixel Count, and Morphological Tools
  - Locate and Measure Circle Tools
  - Blob and Flaw Detection
  - Distance and Pattern Match

Bug Fixes

- `[MVFIS-2256]` Resolved an issue to ensure licensing status remains synchronized and accurate.
- `[MVFIS-2231]` Improved Pharmacode decoding to successfully read symbols with uniform bar widths.
- `[MVFIS-2209]` Increased the stability of the Connectivity Gateway to prevent random device connection loss.
- `[MVFIS-2207]` Addressed an issue in the Deep Learning OCR tool to improve the reliability of character search.
- `[MVFIS-2186]` Resolved an issue to ensure licensing status remains synchronized and accurate.
- `[MVFIS-2160]` Improved device stability to prevent reboots on the FS42 when using a 24V trigger.
- `[MVFIS-2145]` Updated tooltips to provide accurate guidance when the Automation Wedge feature is active.
- `[MVFIS-2111]` Improved FTP connection reliability for FS80 devices.
- `[MVFIS-2018]` Improved decoding performance on the FS10 for challenging labels and DataMatrix codes.
- `[MVFIS-1946]` Updated the Industrial Ethernet/IP EDS file to add support for xS42 devices.
- `[MVFIS-1789]` Corrected an issue where a device would incorrectly report a job as running after a device backup.

- `[MVFIS-2206]` Improved the performance of the WebHMI to provide a real-time display of results and images.

Known Issues and Workarounds

No known issues are reported for this release.

Zebra Aurora Focus™ MV/FIS Configuration Utility This document summarizes the following software release:

Software Version Number	Release Date
V9.4.26	October 2025

Product Description

Zebra Aurora Focus™ is a host-side application which allows for the configuration Zebra FS10, xS20, xS40, xS70, FS42, NS42 and FS80 devices. Zebra Aurora Focus™ Deep Learning Editor is companion software for training the Deep Learning Anomaly Detection tool. To use the editor, a physical or emulated NS42 device licensed for Anomaly Detection is required. It is installed independently and available in two versions: CPU and GPU versions. The GPU version is to train models using NVIDIA GPUs.

Device Compatibility

Compatibility Item	Details
Programming Modes	USB connected cable; M12 Ethernet

Supported Devices	Supported Device FW
(BSP) GS20	**9.4.696 -** CAAGDS00-006-R43
(BSP) FS40, VS40, FS70, VS70	**9.4.696 -** CAAESS00-006-R43
(BSP) FS42, NS42	**9.4.696 -** CAAGGS00-006-R43
(BSP) FS10, FS20, VS20	**9.4.706 -** CAAFFS00-006-R44
(BSP) FS80	**9.4.185 -** CAAHMS00-001-R59

Important Notes

- If downgrading firmware from newer version to an older version, a factory reset through Aurora Focus, or the HMI must be performed after it completes. Failure to do so could result in the device not functioning properly.
- After upgrading the FS80 to the 9.4 release, downgrades can no longer be completed with an incremental update package and must be done with the full upgrade package. To downgrade from 9.4, install the 9.1 Full Update Firmware first, and then install the desired firmware if different than 9.1.
- If you are using an FS10/20 device with version 9.4.696, upgrade to version 9.4.706 at your earliest convenience. Refer to **[MVFIS-2017] FS10** and **[MVFIS-2172]** of the bug fix section for more information.

Components

Components	Version
Aurora Focus	**9.4.26**
Aurora Focus HMI	**10.0.26**
Connectivity Gateway	**1.3.6**
Aurora Deep Learning	**9.4.11193**
(BSP) GS20	**9.4.696 -** CAAGDS00-006-R43
(BSP) FS40, VS40, FS70, VS70	**9.4.696 -** CAAESS00-006-R43
(BSP) FS42, NS42	**9.4.696 -** CAAGGS00-006-R43
(BSP) FS10, FS20, VS20	**9.4.706 -** CAAFFS00-006-R44
(BSP) FS80	**9.4.185 -** CAAHMS00-001-R59

The components are installed in the following folders:

Component	Location
Application	Program Files\Zebra Technologies\Aurora Focus*version*
Job/Configuration Files	Users\Public\Documents\ZebraAurora*ModelType*\Jobs
Saved Images	Users\Public\Documents\ZebraAurora\Captures*xSxxxxxx* Camera

New Features

V9.4.26

RELEASE DATE: October 2025

- Thales Licensing Migration
  - Activate, provision, deactivate and deprovision licenses to devices without leaving Aurora Focus.
  - Manage licenses online or offline using Aurora Focus or the Web HMI.
- Job Slot Management
  - Reuse, assign and re-assign in Device Settings or using TCP/IP commands.
- Reuse a job slot or re-assign the job slot number.
  - Change jobs using communication command.
- Assign jobs to specific slot numbers.
- Support for Pharmacode Symbologies
  - Pharmacode symbology is now available in both the Barcode and the DataCode tool.
- FS80 Enhancements
- Enable the discovery of follower devices in the Connectivity Gateway in Device Settings.
- Support for Leader & Follower Lite functionality for standard configurations without additional licenses.

- The FS80 now supports communication with PLCs using Modbus or Ethernet/IP protocol.
- Deep Learning OCR Enhancements
  - A Minimum Code Quality metric is now returned for each character returned by the OCR tool.
- Filmstrip Management
  - Configure whether or not to capture an image when exiting deployment mode.
- Obtain Device MAC Address Using JavaScript
  - Retrieve the device MAC address using JavaScript
- Zebra Integrated Multifunction Light (ZIML) Support on FS42 5MP
  - Support for using FS42 5MP devices with the ZIML
- Additional Enhancements
  - Aurora Deep Learning Service launches automatically when using the editor.
  - Anomaly Detection UI improvements.
- Support for all trigger modes using Datacode tools.
  - Web HMI UI improvements for performing a Factory Reset.
  - Support for the FS10/20 configurations with plastic exit window.
- MV Essential Toolset is now included in Deep Learning OCR and Deep Learning Anomaly Detection licenses.

Bug Fixes

- **[MVFIS-2044]** The image and heatmap in the ROI of the anomaly detection tool are incorrectly oriented and cropped.
- **[MVFIS-2023]** Pulse width in AF Device settings and ZCG is conflicting.
- **[MVFIS-2118]** FIS devices stop triggering when deploying some JavaScript.
- **[MVFIS-2116]** Focus diopter vs mm strange behavior.
- **[MVFIS-2112]** Trigger ID reset issue.
- **[MVFIS-2138] NS42:** Aurora Focus does not load Jobs to the NS42.
- **[MVFIS-2127] FS42:** Switching back to edit mode blocks the ability to set the focal point in live view mode on FS42.
- **[MVFIS-2098] FS10:** USB HID no output while network is disabled.
- **[MVFIS-2088] FS42:** HID does not re-connect if the client (Windows or Linux) is rebooted while the FS42 is powered by PoE and triggering. Reboot the device to re-establish HID connection.
- **[MVFIS-2054] FS42:** Focus not updating focus correctly.
- **[MVFIS-2039] FS40:** Upgrading from 7.0 to 9.2 breaks the unit.
- **[MVFIS-2020] NS42:** HID output - only works with USB powered device only.
- **[MVFIS-1994] FS40:** During fast triggering on turntable, job stalls (triggers still recognize and the illumination light still flashes).
- **[MVFIS-1879] FS20/40:** Re-reading - Auto (Self), Continuous, Barcode must leave the field of view (FOV).
- **[MVFIS-2017] FS10**: Upgrade from 9.1 to 9.2.7 fails - Lightning/Charger boot problem. Issue resolved with version **9.4.706 -** CAAFFS00-006-R44, which replaces the previous version (**9.4.696 -** CAAFFS00-006-R43).
- **[MVFIS-2172]** Upgrading FS10's firmware from 3.1.40 to 9.3.659 breaks the unit. Issue resolved with version **9.4.706 -** CAAFFS00-006-R44, which replaces the previous version (**9.4.696 -** CAAFFS00-006-R43).

Release Notes (Previous Releases)

V9.3.12

RELEASE DATE: JUNE 2025

V9.2.7

RELEASE DATE: APRIL 2025

V9.1.6

RELEASE DATE: FEBRUARY 2025

V9.0.12

RELEASE DATE: DECEMBER 2024

V7.0.48000

RELEASE DATE: MAY 2024

V6.2.70000

RELEASE DATE: FEBRUARY 2024

V6.1.70000

RELEASE DATE: SEPTEMBER 2023

V6.0.22000

RELEASE DATE: JULY 2023

V5.0.13000

RELEASE DATE: FEBRUARY 2023

V4.0.22000

RELEASE DATE: DECEMBER 2022

V3.1.40000

RELEASE DATE: AUGUST 2022

V3.0.10000

RELEASE DATE: JUNE 2022

V2.0.8000

RELEASE DATE: APRIL 2022

V2.0.7000

RELEASE DATE: MARCH 2022

V1.200.2001

RELEASE DATE: DECEMBER 2021

V1.148.0001

RELEASE DATE: OCTOBER 2021

V1.134.0001 - Initial Product Release

RELEASE DATE: JULY 2021

Minimum PC Requirements

Component	Location
CPU	Intel or AMD 64-bit processor
Random Access Memory Files	4 GB | Recommended: 8GB
GPU	GTX 1030, Intel UHD 620 or equivalent
Graphics Memory	2GB | Recommended: 4GB
Disk Space	4GB free disk space
Monitor Resolution	1280 x 800 | Recommended: 1920 x 1080
Operating System	Windows 10
Internet Connection	Required for software activation

Installation

Aurora Focus™ can be launched from the Windows 10 Start button.

- Start / All Programs / Zebra Scanner / Zebra Aurora Focus <*version*>

Zebra Aurora Focus™ MV/FIS Configuration Utility

Description

Product Description Zebra Aurora Focus™ is a host-side application which allows for the configuration Zebra FS10, xS20, xS40, xS70, FS42, NS42 and FS80 devices. Zebra Aurora Focus™ Deep Learning Editor is companion software for training the Deep Learning Anomaly Detection tool. To use the editor, a physical or emulated NS42 device licensed for Anomaly Detection is required. It is installed independently and available in two versions: CPU and GPU versions. The GPU version is to train models using NVIDIA GPUs.

Device Compatibility

Compatibility Item	Details
Programming Modes	USB connected cable; M12 Ethernet Note:; If downgrading firmware from newer version to an older version, a factory reset through Aurora Focus, or the HMI must be performed after it completes. Failure to do so could result in device not functioning properly.; After upgrading the FS80 to the 9.3 release, downgrades can no longer be completed with an incremental update package and must be done with the full upgrade package. To downgrade from 9.3, install the 9.1 Full Update Firmware first, and then install the desired firmware if different than 9.1.

Supported Devices	Supported Device FW
FS40, VS40, FS70, VS70	9.3.659 - CAAESS00-006-R33
FS42, NS42	9.3.659 - CAAGGS00-006-R33
FS10, FS20, VS20	9.3.659 - CAAFFS00-006-R33
FS80	9.3.159 - CAAHMS00-001-R36

Release Notes

This document summarizes the following software release:

Software Version Number	Release Date
V9.3.12	June 2025

V9.3.12

RELEASE DATE: JUNE 2025

- Profinet Certification for xS20, xS40, xS42, and xS70
  - The 9.3 release has received Profinet certification for the following products xS20, xS40, xS42 and xS70
- Deep Learning OCR Enhancements
  - Expose character quality
- A quality metric is now returned for each character returned by OCR tool. The quality metric is available in JavaScript.
- Expose character height and score
- A score and height are now returned for each character returned by OCR tool. This information is available in JavaScript.
- Support for grammar rules
- Users can now configure grammar rules for the OCR tool, ensuring that the tool only returns results that match the specified pattern.
- Anomaly Detection Enhancements
  - The Anomaly Detection tool now allows the Region of Interest (ROI) to be moved or rotated. Furthermore, the tool can be aligned with other tool based on its results.
  - Deep Learning Editor user experience enhancements
- Support for Connectivity Gateway on the FS80

    - Note that Connectivity Gateway is only available on specific FS80 skus which are not yet available for order. Please contact Product Management and Application Engineering if you have an opportunity that requires this functionality.
- Support for Mailmark Symbology
    - The Mailmark symbology is now available in the barcode tool
- Support for 1D Symbologies in DataCode Tool
    - The DataCode tool now supports several 1D symbologies. This can be used for use cases where the DataCode tool is used to detect 2D DPM codes as well as 1D labels in the same image.
- Additional JavaScript
- Add support for gradient tool in JavaScript
    - Add support for GPIO output control in JavaScript
- Other Enhancements
- Ability to disable ZETI functionality
- The ZETI protocol on port 23 can now either be disabled or moved to another port number. These setting are now available in the Device Settings - Communication page under ZETI settings.
- Connectivity Gateway support for Level start and end control of followers
- Connectivity Gateway now supports stopping a Level Continuous session early after successfully decoding a barcode.

Known Issues

- Under some conditions, barcode decode performance may be impacted.
    - Workarounds:
        - Disable unused symbologies.
        - Disable configuration barcodes (if not used) from the General menu.
        - Set expected module size appropriately.
- On occasion, dual update option on firmware update fails.
    - Workaround: Update firmware without using dual update. Then repeat the same firmware version update using "Force Update" with "Dual Update" checked.
- When `user_data_control` is enabled with appropriate data and data length in AOI, it returns an error code.
    - Workaround: Ignore the error code.
- Sometimes HMI, after removing a license, does not update the screen.
    - Workaround: Refreshing or switching tabs updates the screen.
- Reboot is required to see the Anomaly Detection tool in Aurora Focus after the upgrade license is applied.
- Rarely, it may take a few seconds for the UI to update after selecting or deselecting all symbologies.
- When adding a very large number of tools to the VS flow builder, ROI boundaries may not be visible or performance issues may occur.
- When using Connectivity Gateway and configuring to use more than one GPIO, changes may not take effect immediately.
    - Workaround: Restart Runtime.
- WebHMI objects may not always load properly when opening the WebHMI page.
    - Workaround: Refresh page.
- WebHMI objects may not always load properly on reboot.
    - Workaround: Refresh page.
- If factory resetting the device on a USB connection, or where NTP is not available, the device real-time clock may reset.
    - Workaround: Open WebHMI and navigate to the license page to sync time with the browser host, or set time manually.
- When saving images over FTP using a hostname, transfer delays may occur and the image is not transferred.
    - Workaround: Use FTP server IP address.
- When using Object Plus tools, operation may fail due to contrast value.
    - Workaround: Adjust contrast value until operation succeeds and results are satisfactory.
- Some ZETI commands do not work with the shortened command format.
    - Workaround: Use the extended ZETI command format.
- In some cases, using many image banks causes slight processing delays.
    - Workaround: Reduce number of image banks if possible.
- Device may not be discoverable in Aurora Focus when starting firmware update in HMI until update completes and device reboots.
    - Workaround: Wait for firmware update completion.
- Firmware updates fail if memory consumption is already very high.
    - Workaround: Reboot device before performing firmware update.
- Uninstalling InstallShield does not always remove all registry entries.
    - Workaround: Use `regedit` and manually delete the Aurora Focus folder under `Computer\HKEY_LOCAL_MACHINE\SOFTWARE\Zebra`.

V9.2.7

RELEASE DATE: APRIL 2025

- Automation Wedge (Feature Preview)
    - Support for Automation Wedge (SSI over USB/CDC connection)
- This release includes a feature preview of the Automation Wedge feature which allows Zebra Fixed Industrial Scanning (FIS) smart camera to work with Zebra mobile computer through the Zebra DataWedge mobile computer application.
- Known Limitations:
- Use a USB C to C cable when connecting a camera to a PC/laptop and setting up Automation Wedge. Some PC/laptop machines do not work well when using USB A to C cables and enabling USB SSI.
- Automation Wedge does not currently support multi-code barcode decoding
- JavaScript Enhancements
    - JavaScript support to format and customize HID output
- User can customize output via HID interface and can add special keys into the output for example function keys.
- Support for additional tools in JavaScript
- JavaScript now supports using data from the following tools: Anomaly Detection, Contrast, Edge and OCR
- Support for FTP filename customization via JavaScript
- Tool Enhancements
    - DataCode tool now supports Presentation and Presentation Aggregation Trigger modes
    - DataCode tool now support Continuous and Level Continuous Trigger modes
- FS80 Enhancements
- Support (USB to) Serial port cables
- Use of this feature requires third-party USB to serial cables/dongles
- Other Enhancements
    - Improvements to Match String features
- A different Match String can be used for each tool in a job
- Add option co configure Match Mode (Disabled, Exact, Contain, Regex) in Aurora and using communication channels
- Add option to provide Match String as Base64
- Add full support for Match String to OCR and DataCode tools
    - ZETI Commands:
- match_string_get
- match_string_update [match_mode=STRING_EXACT/DISABLED/STRING_CONTAINS/REGEX"] [persistent=true/false] [format=PLAIN/BASE64] [tool=toolname/friendly name] [value=newstring]
- TCP/IP Connection Test Feature
- The TCP/IP Connection Test feature allows the user to test both the TCP output and TCP control ports to make sure the connection works.

V9.1.6

RELEASE DATE: FEBRUARY 2025

- Support for new FS42 5MP resolution camera
- JavaScript enhancements
    - Support for sending raw bytes in addition to sending text data in results output
    - Serial port output control and sending unique output to the serial port
    - TCP output control and sending unique output to the TCPIP connection
    - Control of job pass fail results
    - Support JavaScript on the FS80
    - Additional device, inspection and performance metric information is now available for use in JavaScript
- Tool enhancements
    - Fixturing from Barcode and Datacode tools
    - Contrast tool improvements
- FS80 enhancements
    - APIPA: Automatic private IP fallback when DCHP fails to assign IP address on FS80

V9.0.12

RELEASE DATE: DECEMBER 2024

- Support for the NS42 Camera
- Support for the FS80 Camera. Note the following will be supported in future releases
    - JavaScript
- FS80 Connectivity Gateway (Leader/Follower), GS20 is required
    - Image banks (Image Perfect)
    - Presentation mode
- Communication with PLCs (Ethernet EIP, Modbus, Profinet)
- ZETI command to control GPIO
    - SW license upgrade capability
    - Configuration barcodes
- Support for Multi-Function Integrated Light for FS80
- NEW tool: Deep Learning Anomaly Detection
- Deep Learning Editor (for training the Anomaly Detection tool)
- JavaScript Support
    - Output formatting
- Output over TCP, Serial Port, CDC and basic HID (function keys not supported)
    - Editor with IntelliSense
    - Console logging for basic debugging
- Support for explicit serial trigger command
- Support for Multiple Aurora Focus versions installed on same host
- Option to select USB2 or USB3 for xS40/xS42/xS70
- Support for NPU-accelerated Deep Learning OCR
- Support for AutoTune on DPM with the Datacode tool
- Support for DPM QR code reading
- Support for reading QR Codes in Datacode tool
- Lower selectable heartbeat intervals
- Support for aggregating codes in presentation trigger mode - Multiple Aurora Focus stability changes and enhancements
- Miscellaneous bug fixes
- Improved debugging for trigger overrun conditions when enabling logging on the camera

V7.0.48000

RELEASE DATE: MAY 2024

- When editing a job for a camera, editing occurs directly on device and then timing values are more realistic.
- Support for the new FS42 camera
- Support for Separate (Explicit) TCP start and stop trigger strings - Support for TCP control and outport on same port number.
- Connectivity Gateway: Support for backup and restore for configuration (HMI and ZETI)
- Connectivity Gateway: Support for level trigger Mode
- Connectivity Gateway: Separate start and stop TCP/Serial triggers
- Connectivity Gateway: Output at end of level trigger mode
- Connectivity Gateway: UI/UX Improvements
    - Now include helper tips in each section
    - Debugging tools added to bottom of Setup screen - DL-OCR Match string settable by TCP command via ZETI
- Debug mode to save images in both level and continuous modes
- Support for saving images based on a time interval
- ZETI support for ASCII format when sending match string commands
- Multiple performance improvements
- Multiple Aurora Focus stability changes and enhancements
- Miscellaneous Bug Fixes

V6.2.70000

RELEASE DATE: FEBRUARY 2024

- Support for Liquid Lens focal distance using millimeters or inches.
- 2D Configuration barcode support to perform Autotune (Focus only)
- ZETI Command support to perform Autotune (Focus only)

V6.1.70000

RELEASE DATE: SEPTEMBER 2023

- Performance and Enhancements
    - Datacode Tool Improvements and Optimizations
    - Quantized Deep Learning OCR
    - FTP/SFTP folder storage suffix options
    - FTP/SFTP option to disable JSON results file
- ADF Output Enhancements
    - Support for Function Keys: F1-12
    - Support for Modifier Keys CTRL and ALT
- Option to Disable Tune and Trigger hardware buttons
- ZETI Interface improvements
- Backup and Restore feature improvements
- Miscellaneous Bug Fixes

V6.0.22000

RELEASE DATE: JULY 2023

- 2D Configuration Barcode Generation and Deployment
- Advanced Filtering and Formatting of Decode Data
- Multi-device management (Up to 5 devices at a time)
    - Firmware Update
    - Backup/Restore
    - Device Log Download
- Zebra Connectivity Gateway for Leader/Followers scanner configuration
- ZETI Command Interface Additional Command Support
- Aurora Focus UX Improvements
    - Improvements to Custom Formatting
    - Improved Localization
- Performance and Enhancements
    - Decode performance improvements.
    - Enhanced Session Based Triggering with Image Save
    - Additional Localization Enhancements

V5.0.13000

RELEASE DATE: FEBRUARY 2023

- Technology Previews
    - Aurora Focus and HMI Language Support - Simplified Chinese, Korean, German, French and Spanish
    - Expanded ZETI Interface Commands
- Tools
    - Deep Learning Based OCR for Fixed Industrial Scanners
- Fully Released - Activated via Licenses
- Performance

    - Improvements to deployed job run consistency including Barcode tool timeout, decoding, and trigger timing consistency.
    - Level Trigger Enhancements
    - Triggered Image Buffering
- UX Improvements
    - Backup/Restore/Firmware Upgrade in Aurora Focus
- Device Configuration 1D Parameter Barcode Setup

V4.0.22000

RELEASE DATE: DECEMBER 2022

- Added Measure Object Tool
- Added Flaw Detection Edge and Intensity Tools
- Added Object Locate Counting
- Added Image Filters: Gradient, Gradient Horizontal, Vertical and Full
- MV Output Formatting on Supported Interfaces (Global or Interface Specific)
- Added Find and Replace to Barcode Decode Tool
- Improved 1D/2D Barcode Decode Performance
- Setting Barcode Match String via TCP or Serial input
- Added Image Download Capability via HMI
- Enhanced Session Based Triggering for Level, Continuous and Presentation Triggers w/Exit Criteria/Aggregation
- Technology Preview: Device Backup/Restore Functionality
- Performance and Stability Enhancements

V3.1.40000

RELEASE DATE: AUGUST 2022

- Added Reset on Trigger for Do Note Read Same Barcode
- Spanish HID Keyboard support
- Performance and Stability Enhancements

V3.0.10000

RELEASE DATE: JUNE 2022

- Added Edge Counting
- Added Deep-Learning Based OCR Tool (in Beta)
- Added Circle Find Tool
- Added Circle Diameter Measurement Tool
- Added a Circle ROI Type
- Added Open and Close Filters
- Added Erode Filter
- Added RS-232 Input Trigger
- Additional 2D Barcode Performance Improvements
- Added ADF to include Pass/Fail Data Types
- Added ADF to include Barcode Quality Metrics Data
- Performance and Stability Enhancements

V2.0.8000

RELEASE DATE: APRIL 2022

- Performance and Stability Enhancements

V2.0.7000

RELEASE DATE: MARCH 2022

- Performance and Stability Enhancements
- 1D Barcode Quality Metrics
- Added Object Locate Plus
- Added Blob Tools
- Added Binarize Filter/Tool
- Added Dilate Filter/Tool
- ImagePerfect(+) fixes & improvements
- Added Scale Calibration for Distance tool
- Added Advanced Pass/Fail Configuration (AND/OR logic for Job results)
- HID Keyboard Expansion - added French and German KB
- Added support for Codabar symbology
- Added Encoding type (Text/Base64) for ADF Full String data type and added option to show nonprintable characters as hex values
- For xS10/20, added external illumination support
- Added "File Path" option when using (S)FTP to Save images
- Added Ability to Open a Local Job (created with emulator) On a Real Connected Camera

V1.200.2001

RELEASE DATE: DECEMBER 2021

- Performance and Stability Enhancements
- Support for 5MP xS40 and xS70 models
- ImagePerfect(+) for FS models
- Option to output Barcode Location Co-ordinates
- Option to not read last n number of barcodes
- Device can be a TCP/IP Client
- Configurable RS-232 settings
- Ability to configure GPIO Trigger delay
- Support for Fast Hardware Trigger (2.3 MP models only)
- Ability to set GPIOs to static high or low
- Added Job controls to clone, rename and delete jobs
- Support for Conditional Regular expressions for Barcode String Match
- Barcode String Match Regular expression support for ManyCode
- Backwards compatibility of Job files
- Decode and MV statistics in Aurora Focus and HMI stored and queried from device

V1.148.0001

RELEASE DATE: OCTOBER 2021

- Performance Enhancements
- Stability Improvements
- Implementation of licensing error messages in Web HMI
- Fixed issue of Job pausing on refresh of Web HMI
- Fixed issue of crash when clicking on "Communications" tab
- Fixed inability to "View" active Jobs from Job menu when Job utilizes a trigger that is automatic in nature.
- Fixed issue of slow decode rate when "Fast 1D/2D" license is enabled
- Added Configurable TCP/IP - results port and trigger string can be configured
- Support for GPIO > Hardware trigger (normal speed)
- Allow setting a job as Startup / Power up job
- Emulators now have license level
- Supports a total of 21 emulators

- ZJOB backwards compatibility and auto conversion of old job files (Windows only) - Seamless (live) tool changes when Upgrade licenses are applied

V1.134.0001 - Initial Product Release

RELEASE DATE: JULY 2021

- Bug Fixes
- Stability Improvements

Components

SW Component	Version
InstallShield	9.3.12
Aurora Focus™	9.3.12
Aurora Focus Deep Learning Editor	9.3.10814
BSP (xS40/70)	9.3.659 - CAAESS00-006-R33
BSP (FS42/NS42)	9.3.659 - CAAGGS00-006-R33
BSP (FS10/xS20)	9.3.659 - CAAFFS00-006-R33
BSP (FS80)	9.3.159 - SAAHMS00-001-R36
Aurora Focus™ HMI	10.0.17
Connectivity Gateway	1.3.3

The components are installed in the following folders:

Component	Location
Application	Program Files\Zebra Technologies\Aurora Focus\version
Job/Configuration Files	Users\Public\Documents\ZebraAurora\ModelType\Jobs
Saved Images	Users\Public\Documents\ZebraAurora\Captures\xSxxxxxx Camera

Minimum PC Requirements

Component	Location
CPU	Intel or AMD 64-bit processor
Random Access Memory Files	4 GB | Recommended: 8GB
GPU	GTX 1030, Intel UHD 620 or equivalent
Graphics Memory	2GB | Recommended: 4GB
Disk Space	4GB free disk space
Monitor Resolution	1280 x 800 | Recommended: 1920 x 1080
Operating System	Windows 10
Internet Connection	Required for software activation

Installation

Aurora Focus™ can be launched from the Windows 10 Start button.

- Start / All Programs / Zebra Scanner / Zebra Aurora Focus <version>

Zebra Aurora Focus™ MV/FIS Configuration Utility

Description

Product Description Zebra Aurora Focus™ is a host-side application which allows for the configuration Zebra FS10, xS20, xS40, xS70, FS42, NS42 and FS80 devices. Zebra Aurora Focus™ Deep Learning Editor is companion software for training the Deep Learning Anomaly Detection tool. To use the editor, a physical or emulated NS42 device licensed for Anomaly Detection is required. It is installed independently and available in two versions: CPU and GPU versions. The GPU version is to train models using NVIDIA GPUs.

Device Compatibility

Compatibility Item	Details
Programming Modes	USB connected cable; M12 Ethernet

Supported Devices	Supported Device FW
FS40, VS40, FS70, VS70	**9.2.631 -** CAAESS00-006-R16
FS42, NS42	**9.2.631 -** CAAGGS00-006-R16
FS10, FS20, VS20	**9.2.631 -** CAAFFS00-006-R16
FS80	**9.2.139 -** SAAHMS00-001-R19

Release Notes

This document summarizes the following software release:

Software Version Number	Release Date
V9.2.7	April 2025

V9.2.7

RELEASE DATE: APRIL 2025

- Automation Wedge (Feature Preview)
  - Support for Automation Wedge (SSI over USB/CDC connection)
- This release includes a feature preview of the Automation Wedge feature which allows Zebra Fixed Industrial Scanning (FIS) smart camera to work with Zebra mobile computer through the Zebra DataWedge mobile computer application.
- Known Limitations:
- Use a USB C to C cable when connecting a camera to a PC/laptop and setting up Automation Wedge. Some PC/laptop machines do not work well when using USB A to C cables and enabling USB SSI.
- Automation Wedge does not currently support multi-code barcode decoding
- JavaScript Enhancements
  - JavaScript support to format and customize HID output User can customize output via HID interface and can add special keys into the output (for example, function keys).
- Support for additional tools in JavaScript
- JavaScript now supports using data from the following tools: Anomaly Detection, Contrast, Edge and OCR
- Support for FTP filename customization via JavaScript
- Tool Enhancements
  - DataCode tool now supports Presentation and Presentation Aggregation Trigger modes
  - DataCode tool now support Continuous and Level Continuous Trigger modes
- FS80 Enhancements
- Support (USB to) Serial port cables
- Use of this feature requires third-party USB to serial cables/dongles
- Other Enhancements
  - Improvements to Match String features
- A different Match String can be used for each tool in a job
- Add option co configure Match Mode (Disabled, Exact, Contain, Regex) in Aurora and using communication channels
- Add option to provide Match String as Base64
- Add full support for Match String to OCR and DataCode tools
  - ZETI Commands:
- match_string_get
- match_string_update [match_mode=STRING_EXACT/DISABLED/STRING_CONTAINS/REGEX"] [persistent=true/false] [format=PLAIN/BASE64] [tool=toolname/friendly name] [value=newstring]
- TCP/IP Connection Test Feature

- The TCP/IP Connection Test feature allows the user to test both the TCP output and TCP control ports to make sure the connection works.

Known Issues

- Under some conditions, barcode decode performance may be impacted.
  - Workarounds:
    - Disable unused symbologies.
    - Disable configuration barcodes (if not used) from the General menu.
    - Set expected module size appropriately.
- On occasion, dual update option on firmware update fails.
  - Workaround: Update firmware without using dual update. Then repeat the same firmware version update using "Force Update" with "Dual Update" checked.
- "Test Connection" can fail with currently used/open device ports.
  - Workaround: Test connection before ports are selected and/or opened during job deployment.
- Unrestricted USB-A Power settings are not transferable with Backup/Restore.
  - Workaround: Use configuration barcodes where unrestricted USB-A Power settings must be cloned.
- When `user_data_control` is enabled with appropriate data and data length in AOI, it returns an error code.
  - Workaround: Ignore the error code.
- Sometimes HMI, after removing a license, does not update the screen.
  - Workaround: Refreshing or switching tabs updates the screen.
- Reboot is required to see the Anomaly Detection tool in Aurora Focus after the upgrade license is applied.
- Rarely, it may take a few seconds for the UI to update after selecting or deselecting all symbologies.
- When adding a very large number of tools to the VS flow builder, ROI boundaries may not be visible or performance issues may occur.
- When using Connectivity Gateway and configuring to use more than one GPIO, changes may not take effect immediately.
  - Workaround: Restart Runtime.
- WebHMI objects may not always load properly when opening the WebHMI page.
  - Workaround: Refresh page.
- WebHMI objects may not always load properly on reboot.
  - Workaround: Refresh page.
- If factory resetting the device on a USB connection, or where NTP is not available, the device real-time clock may reset.
  - Workaround: Open WebHMI and navigate to the license page to sync time with the browser host, or set time manually.
- When saving images over FTP using a hostname, transfer delays may occur and the image is not transferred.
  - Workaround: Use FTP server IP address.
- When using Object Plus tools, operation may fail due to contrast value.
  - Workaround: Adjust contrast value until operation succeeds and results are satisfactory.
- Some ZETI commands do not work with the shortened command format.
  - Workaround: Use the extended ZETI command format.
- In some cases, using many image banks causes slight processing delays.
  - Workaround: Reduce number of image banks if possible.
- Device may not be discoverable in Aurora Focus when starting firmware update in HMI until update completes and device reboots.
  - Workaround: Wait for firmware update completion.
- Firmware updates fail if memory consumption is already very high.
  - Workaround: Reboot device before performing firmware update.
- Uninstalling InstallShield does not always remove all registry entries.
  - Workaround: Use `regedit` and manually delete the Aurora Focus folder under `Computer\HKEY_LOCAL_MACHINE\SOFTWARE\Zebra`.

V9.1.6

RELEASE DATE: FEBRUARY 2025

- Support for new FS42 5MP resolution camera
- JavaScript enhancements
  - Support for sending raw bytes in addition to sending text data in results output
  - Serial port output control and sending unique output to the serial port
  - TCP output control and sending unique output to the TCPIP connection

    - Control of job pass fail results
    - Support JavaScript on the FS80
    - Additional device, inspection and performance metric information is now available for use in JavaScript
- Tool enhancements
    - Fixturing from Barcode and Datacode tools
    - Contrast tool improvements
- FS80 enhancements
    - APIPA: Automatic private IP fallback when DCHP fails to assign IP address on FS80

V9.0.12

RELEASE DATE: DECEMBER 2024

- Support for the NS42 Camera
- Support for the FS80 Camera. Note the following will be supported in future releases
    - JavaScript
    - FS80 Connectivity Gateway (Leader/Follower), GS20 is required
    - Image banks (Image Perfect)
    - Presentation mode
    - Communication with PLCs (Ethernet EIP, Modbus, Profinet)
    - ZETI command to control GPIO
    - SW license upgrade capability
    - Configuration barcodes
- Support for Multi-Function Integrated Light for FS80
- NEW tool: Deep Learning Anomaly Detection
- Deep Learning Editor (for training the Anomaly Detection tool)
- JavaScript Support
    - Output formatting
    - Output over TCP, Serial Port, CDC and basic HID (function keys not supported)
    - Editor with IntelliSense
    - Console logging for basic debugging
- Support for explicit serial trigger command
- Support for Multiple Aurora Focus versions installed on same host
- Option to select USB2 or USB3 for xS40/xS42/xS70
- Support for NPU-accelerated Deep Learning OCR
- Support for AutoTune on DPM with the Datacode tool
- Support for DPM QR code reading
- Support for reading QR Codes in Datacode tool
- Lower selectable heartbeat intervals
- Support for aggregating codes in presentation trigger mode
- Multiple Aurora Focus stability changes and enhancements
- Miscellaneous bug fixes
- Improved debugging for trigger overrun conditions when enabling logging on the camera

V7.0.48000

RELEASE DATE: MAY 2024

- When editing a job for a camera, editing occurs directly on device and then timing values are more realistic.
- Support for the new FS42 camera
- Support for Separate (Explicit) TCP start and stop trigger strings
- Support for TCP control and outport on same port number.
- Connectivity Gateway: Support for backup and restore for configuration (HMI and ZETI)
- Connectivity Gateway: Support for level trigger Mode
- Connectivity Gateway: Separate start and stop TCP/Serial triggers
- Connectivity Gateway: Output at end of level trigger mode
- Connectivity Gateway: UI/UX Improvements
    - Now include helper tips in each section
    - Debugging tools added to bottom of Setup screen DL-OCR Match string settable by TCP command via ZETI Debug mode to save images in both level and continuous modes > > Support for saving images based on a

time interval ZETI support for ASCII format when sending match string commands Multiple performance improvements > > Multiple Aurora Focus stability changes and enhancements Miscellaneous Bug Fixes

V6.2.70000

RELEASE DATE: FEBRUARY 2024

- Support for Liquid Lens focal distance using millimeters or inches.
- 2D Configuration barcode support to perform Autotune (Focus only)
- ZETI Command support to perform Autotune (Focus only)

V6.1.70000

RELEASE DATE: SEPTEMBER 2023

- Performance and Enhancements
    - Datacode Tool Improvements and Optimizations
    - Quantized Deep Learning OCR
    - FTP/SFTP folder storage suffix options
    - FTP/SFTP option to disable JSON results file
- ADF Output Enhancements
    - Support for Function Keys: F1-12
    - Support for Modifier Keys CTRL and ALT
- Option to Disable Tune and Trigger hardware buttons
- ZETI Interface improvements
- Backup and Restore feature improvements
- Miscellaneous Bug Fixes

V6.0.22000

RELEASE DATE: JULY 2023

- 2D Configuration Barcode Generation and Deployment
- Advanced Filtering and Formatting of Decode Data
- Multi-device management (Up to 5 devices at a time)
    - Firmware Update
    - Backup/Restore
    - Device Log Download Zebra Connectivity Gateway for Leader/Followers scanner configuration > > ZETI Command Interface Additional Command Support > > Aurora Focus UX Improvements
    - Improvements to Custom Formatting
    - Improved Localization
- Performance and Enhancements
    - Decode performance improvements.
    - Enhanced Session Based Triggering with Image Save
    - Additional Localization Enhancements

V5.0.13000

RELEASE DATE: FEBRUARY 2023

- Technology Previews
    - Aurora Focus and HMI Language Support - Simplified Chinese, Korean, German, French and Spanish
    - Expanded ZETI Interface Commands
- Tools
    - Deep Learning Based OCR for Fixed Industrial Scanners - Fully Released and Activated via Licenses
- Performance

    - Improvements to deployed job run consistency including Barcode tool timeout, decoding, and trigger timing consistency.
    - Level Trigger Enhancements
    - Triggered Image Buffering
- UX Improvements
    - Backup/Restore/Firmware Upgrade in Aurora Focus
- Device Configuration 1D Parameter Barcode Setup

V4.0.22000

RELEASE DATE: DECEMBER 2022

Added Measure Object Tool > > Added Flaw Detection Edge and Intensity Tools > > Added Object Locate Counting > > Added Image Filters: Gradient, Gradient Horizontal, Vertical and Full > > MV Output Formatting on Supported Interfaces (Global or Interface Specific) > > Added Find and Replace to Barcode Decode Tool > > Improved 1D/2D Barcode Decode Performance

- Setting Barcode Match String via TCP or Serial input
- Added Image Download Capability via HMI
- Enhanced Session Based Triggering for Level, Continuous and Presentation Triggers w/Exit Criteria/Aggregation
- Technology Preview: Device Backup/Restore Functionality
- Performance and Stability Enhancements

V3.1.40000

RELEASE DATE: AUGUST 2022

- Added Reset on Trigger for Do Note Read Same Barcode
- Spanish HID Keyboard support
- Performance and Stability Enhancements

V3.0.10000

RELEASE DATE: JUNE 2022

- Added Edge Counting
- Added Deep-Learning Based OCR Tool (*in Beta*)
- Added Circle Find Tool
- Added Circle Diameter Measurement Tool
- Added a Circle ROI Type
- Added Open and Close Filters
- Added Erode Filter
- Added RS-232 Input Trigger
- Additional 2D Barcode Performance Improvements
- Added ADF to include Pass/Fail Data Types
- Added ADF to include Barcode Quality Metrics Data
- Performance and Stability Enhancements

V2.0.8000

RELEASE DATE: APRIL 2022

- Performance and Stability Enhancements

V2.0.7000

RELEASE DATE: MARCH 2022

- Performance and Stability Enhancements
- 1D Barcode Quality Metrics
- Added Object Locate Plus
- Added Blob Tools
- Added Binarize Filter/Tool
- Added Dilate Filter/Tool
- ImagePerfect(+) fixes & improvements
- Added Scale Calibration for Distance tool
- Added Advanced Pass/Fail Configuration (AND/OR logic for Job results)
- HID Keyboard Expansion - added French and German KB
- Added support for Codabar symbology
- Added Encoding type (Text/Base64) for ADF Full String data type and added option to show nonprintable characters as hex values
- For xS10/20, added external illumination support
- Added "File Path" option when using (S)FTP to Save images
- Added Ability to Open a Local Job (created with emulator) On a Real Connected Camera

V1.200.2001

RELEASE DATE: DECEMBER 2021

- Performance and Stability Enhancements
- Support for 5MP xS40 and xS70 models
- ImagePerfect(+) for FS models
- Option to output Barcode Location Co-ordinates
- Option to not read last n number of barcodes
- Device can be a TCP/IP Client Configurable RS-232 settings > > Ability to configure GPIO Trigger delay > > Support for Fast Hardware Trigger (2.3 MP models only) > > Ability to set GPIOs to static high or low
- Added Job controls to clone, rename and delete jobs
- Support for Conditional Regular expressions for Barcode String Match
- Barcode String Match Regular expression support for ManyCode
- Backwards compatibility of Job files
- Decode and MV statistics in Aurora Focus and HMI stored and queried from device

V1.148.0001

RELEASE DATE: OCTOBER 2021

- Performance Enhancements
- Stability Improvements
- Implementation of licensing error messages in Web HMI
- Fixed issue of Job pausing on refresh of Web HMI
- Fixed issue of crash when clicking on "Communications" tab
- Fixed inability to "View" active Jobs from Job menu when Job utilizes a trigger that is automatic in nature.
- Fixed issue of slow decode rate when "Fast 1D/2D" license is enabled
- Added Configurable TCP/IP - results port and trigger string can be configured
- Support for GPIO > Hardware trigger (normal speed)
- Allow setting a job as Startup / Power up job
- Emulators now have license level
- Supports a total of 21 emulators
- ZJOB backwards compatibility and auto conversion of old job files (Windows only) - Seamless (live) tool changes when Upgrade licenses are applied

V1.134.0001 - Initial Product Release

RELEASE DATE: JULY 2021

- Bug Fixes
- Stability Improvements

Components

SW Component	Version
InstallShield	9.2.7
Aurora Focus™	9.2.7
Aurora Focus Deep Learning Editor	9.0.10631
BSP (xS40/70)	9.2.631 - CAAESS00-006-R16
BSP (FS42/NS42)	9.2.631 - CAAGGS00-006-R16
BSP (FS10/xS20)	9.2.631 - CAAFFS00-006-R16
BSP (FS80)	9.2.139 - SAAHMS00-001-R19
Aurora Focus™ HMI	10.0.11
Connectivity Gateway	1.2.10

The components are installed in the following folders:

Component	Location
Application	Program Files\Zebra Technologies\Aurora Focus/version
Job/Configuration Files	Users\Public\Documents\ZebraAurora/ModelType\Jobs
Saved Images	Users\Public\Documents\ZebraAurora/Captures/xSxxxxxx Camera

Minimum PC Requirements

Component	Location
CPU	Intel or AMD 64-bit processor
Random Access Memory Files	4 GB | Recommended: 8GB
GPU	GTX 1030, Intel UHD 620 or equivalent
Graphics Memory	2GB | Recommended: 4GB
Disk Space	4GB free disk space
Monitor Resolution	1280 x 800 | Recommended: 1920 x 1080
Operating System	Windows 10
Internet Connection	Required for software activation

Installation

Aurora Focus™ can be launched from the Windows 10 Start button.

- Start / All Programs / Zebra Scanner / Zebra Aurora Focus

Zebra Aurora Focus™ MV/FIS Configuration Utility

Description

Product Description Zebra Aurora Focus™ is a host-side application which allows for the configuration Zebra FS10, xS20, xS40, xS70, FS42, NS42 and FS80 devices. Zebra Aurora Focus™ Deep Learning Editor is companion software for training the Deep Learning Anomaly Detection tool. To use the editor, a physical or emulated NS42 device licensed for Anomaly Detection is required. It is installed independently and available in two versions: CPU and GPU versions. The GPU version is to train models using NVIDIA GPUs.

Device Compatibility

Compatibility Item	Details
Programming Modes	USB connected cable; M12 Ethernet

Supported Devices	Supported Device FW
FS40, VS40, FS70, VS70	**9.1.6 -** CAAESS00-006-R08
FS42, NS42	**9.1.6 -** CAAGGS00-006-R08
FS10, FS20, VS20	**9.1.6 -** CAAFFS00-006-R08
FS80	**9.1.7 -** SAAHMS00-001-R10

Release Notes

This document summarizes the following software release:

Software Version Number	Release Date
V9.1.6	February 2025

V9.1.6

RELEASE DATE: FEBRUARY 2025

- Support for new FS42 5MP resolution camera
- JavaScript enhancements
  - Support for sending raw bytes in addition to sending text data in results output
  - Serial port output control and sending unique output to the serial port
  - TCP output control and sending unique output to the TCPIP connection
  - Control of job pass fail results
  - Support JavaScript on the FS80
  - Additional device, inspection and performance metric information is now available for use in JavaScript
- Tool enhancements
  - Fixturing from Barcode and Datacode tools
  - Contrast tool improvements
- FS80 enhancements
  - APIPA: Automatic private IP fallback when DCHP fails to assign IP address on FS80

Known Issues

- Under some conditions, barcode decode performance may be impacted.
  - Workarounds:
    - Disable unused symbologies.
    - Disable configuration barcodes (if not used) from the General menu.
    - Set expected module size appropriately.
- On occasion, dual update option on firmware update fails.
  - Workaround: Update firmware without using dual update. Then repeat the same firmware version update using "Force Update" with "Dual Update" checked.
- String match always fails with "Exact match" if "Expand String Match to Entire Payload" is enabled when using the Datacode tool.
- IE decode field is not sent to PLC when using the Datacode tool.
  - Workaround: Use the Barcode tool.
- When `user_data_control` is enabled with appropriate data and data length in AOI, it returns an error code.
  - Workaround: Ignore the error code.
- Sometimes HMI, after removing a license, does not update the screen.
  - Workaround: Refreshing or switching tabs updates the screen.

- Reboot is required to see the Anomaly Detection tool in Aurora Focus after the upgrade license is applied.
- Rarely, the job list in Aurora Focus does not show a job after reboot.
  - Workaround: Click Refresh at the top of the job list.
- Rarely, it may take a few seconds for the UI to update after selecting or deselecting all symbologies.
- When adding a very large number of tools to the VS flow builder, ROI boundaries may not be visible or performance issues may occur.
- When using Connectivity Gateway and configuring to use more than one GPIO, changes may not take effect immediately.
  - Workaround: Restart Runtime.
- WebHMI objects may not always load properly when opening the WebHMI page.
  - Workaround: Refresh page.
- WebHMI objects may not always load properly on reboot.
  - Workaround: Refresh page.
- If factory resetting the device on a USB connection, or where NTP is not available, the device real-time clock may reset.
  - Workaround: Open WebHMI and navigate to the license page to sync time with the browser host, or set time manually.
- When saving images over FTP using a hostname, transfer delays may occur and the image is not transferred.
  - Workaround: Use FTP server IP address.
- When using Object Plus tools, operation may fail due to contrast value.
  - Workaround: Adjust contrast value until operation succeeds and results are satisfactory.
- Some ZETI commands do not work with the shortened command format.
  - Workaround: Use the extended ZETI command format.
- In some cases, using many image banks causes slight processing delays.
  - Workaround: Reduce number of image banks if possible.
- Device may not be discoverable in Aurora Focus when starting firmware update in HMI until update completes and device reboots.
  - Workaround: Wait for firmware update completion.
- Firmware updates fail if memory consumption is already very high.
  - Workaround: Reboot device before performing firmware update.
- Uninstalling InstallShield does not always remove all registry entries.
  - Workaround: Use `regedit` and manually delete the Aurora Focus folder under `Computer\HKEY_LOCAL_MACHINE\SOFTWARE\Zebra`.

V9.0.12

RELEASE DATE: DECEMBER 2024

- Support for the NS42 Camera
- Support for the FS80 Camera. Note the following will be supported in future releases
  - JavaScript
  - FS80 Connectivity Gateway (Leader/Follower), GS20 is required
  - Image banks (Image Perfect)
  - Presentation mode
  - Communication with PLCs (Ethernet EIP, Modbus, Profinet)
  - ZETI command to control GPIO
  - SW license upgrade capability
  - Configuration barcodes
- Support for Multi-Function Integrated Light for FS80
- NEW tool: Deep Learning Anomaly Detection
- Deep Learning Editor (for training the Anomaly Detection tool)
- JavaScript Support
  - Output formatting
  - Output over TCP, Serial Port, CDC and basic HID (function keys not supported)
  - Editor with IntelliSense
  - Console logging for basic debugging
- Support for explicit serial trigger command
- Support for Multiple Aurora Focus versions installed on same host
- Option to select USB2 or USB3 for xS40/xS42/xS70
- Support for NPU-accelerated Deep Learning OCR
- Support for AutoTune on DPM with the Datacode tool
- Support for DPM QR code reading
- Support for reading QR Codes in Datacode tool

- Lower selectable heartbeat intervals
- Support for aggregating codes in presentation trigger mode
- Multiple Aurora Focus stability changes and enhancements
- Miscellaneous bug fixes
- Improved debugging for trigger overrun conditions when enabling logging on the camera

V7.0.48000

RELEASE DATE: MAY 2024

- When editing a job for a camera, editing occurs directly on device and then timing values are more realistic.
- Support for the new FS42 camera
- Support for Separate (Explicit) TCP start and stop trigger strings
- Support for TCP control and outport on same port number.
- Connectivity Gateway: Support for backup and restore for configuration (HMI and ZETI)
- Connectivity Gateway: Support for level trigger Mode
- Connectivity Gateway: Separate start and stop TCP/Serial triggers
- Connectivity Gateway: Output at end of level trigger mode
- Connectivity Gateway: UI/UX Improvements
    - Now include helper tips in each section
    - Debugging tools added to bottom of Setup screen
- DL-OCR Match string settable by TCP command via ZETI
- Debug mode to save images in both level and continuous modes
- Support for saving images based on a time interval
- ZETI support for ASCII format when sending match string commands
- Multiple performance improvements
- Multiple Aurora Focus stability changes and enhancements
- Miscellaneous Bug Fixes

V6.2.70000

RELEASE DATE: FEBRUARY 2024

- Support for Liquid Lens focal distance using millimeters or inches.
- 2D Configuration barcode support to perform Autotune (Focus only)
- ZETI Command support to perform Autotune (Focus only)

V6.1.70000

RELEASE DATE: SEPTEMBER 2023

- Performance and Enhancements
    - Datacode Tool Improvements and Optimizations
    - Quantized Deep Learning OCR
    - FTP/SFTP folder storage suffix options
    - FTP/SFTP option to disable JSON results file
- ADF Output Enhancements
    - Support for Function Keys: F1-12
    - Support for Modifier Keys CTRL and ALT
- Option to Disable Tune and Trigger hardware buttons
- ZETI Interface improvements
- Backup and Restore feature improvements
- Miscellaneous Bug Fixes

V6.0.22000

RELEASE DATE: JULY 2023

- 2D Configuration Barcode Generation and Deployment
- Advanced Filtering and Formatting of Decode Data
- Multi-device management (Up to 5 devices at a time)
    - Firmware Update
    - Backup/Restore
    - Device Log Download
- Zebra Connectivity Gateway for Leader/Followers scanner configuration
- ZETI Command Interface Additional Command Support
- Aurora Focus UX Improvements
    - Improvements to Custom Formatting
    - Improved Localization
- Performance and Enhancements
    - Decode performance improvements.
    - Enhanced Session Based Triggering with Image Save
    - Additional Localization Enhancements

V5.0.13000

RELEASE DATE: FEBRUARY 2023

- Technology Previews
    - Aurora Focus and HMI Language Support - Simplified Chinese, Korean, German, French and Spanish
    - Expanded ZETI Interface Commands
- Tools
    - Deep Learning Based OCR for Fixed Industrial Scanners - Fully Released and Activated via Licenses
- Performance
    - Improvements to deployed job run consistency including Barcode tool timeout, decoding, and trigger timing consistency.
    - Level Trigger Enhancements
    - Triggered Image Buffering
- UX Improvements
    - Backup/Restore/Firmware Upgrade in Aurora Focus
- Device Configuration 1D Parameter Barcode Setup

V4.0.22000

RELEASE DATE: DECEMBER 2022

- Added Measure Object Tool
- Added Flaw Detection Edge and Intensity Tools
- Added Object Locate Counting
- Added Image Filters: Gradient, Gradient Horizontal, Vertical and Full
- MV Output Formatting on Supported Interfaces (Global or Interface Specific)
- Added Find and Replace to Barcode Decode Tool
- Improved 1D/2D Barcode Decode Performance
- Setting Barcode Match String via TCP or Serial input
- Added Image Download Capability via HMI
- Enhanced Session Based Triggering for Level, Continuous and Presentation Triggers w/Exit Criteria/Aggregation
- Technology Preview: Device Backup/Restore Functionality
- Performance and Stability Enhancements

V3.1.40000

RELEASE DATE: AUGUST 2022

- Added Reset on Trigger for Do Note Read Same Barcode
- Spanish HID Keyboard support
- Performance and Stability Enhancements

V3.0.10000

RELEASE DATE: JUNE 2022

- Added Edge Counting
- Added Deep-Learning Based OCR Tool (*in Beta*)
- Added Circle Find Tool
- Added Circle Diameter Measurement Tool
- Added a Circle ROI Type
- Added Open and Close Filters
- Added Erode Filter
- Added RS-232 Input Trigger
- Additional 2D Barcode Performance Improvements
- Added ADF to include Pass/Fail Data Types
- Added ADF to include Barcode Quality Metrics Data
- Performance and Stability Enhancements

V2.0.8000

RELEASE DATE: APRIL 2022

- Performance and Stability Enhancements

V2.0.7000

RELEASE DATE: MARCH 2022

- Performance and Stability Enhancements
- 1D Barcode Quality Metrics
- Added Object Locate Plus
- Added Blob Tools
- Added Binarize Filter/Tool
- Added Dilate Filter/Tool
- ImagePerfect(+) fixes & improvements
- Added Scale Calibration for Distance tool
- Added Advanced Pass/Fail Configuration (AND/OR logic for Job results)
- HID Keyboard Expansion - added French and German KB
- Added support for Codabar symbology
- Added Encoding type (Text/Base64) for ADF Full String data type and added option to show nonprintable characters as hex values
- For xS10/20, added external illumination support
- Added "File Path" option when using (S)FTP to Save images
- Added Ability to Open a Local Job (created with emulator) On a Real Connected Camera

V1.200.2001

RELEASE DATE: DECEMBER 2021

- Performance and Stability Enhancements
- Support for 5MP xS40 and xS70 models
- ImagePerfect(+) for FS models
- Option to output Barcode Location Co-ordinates
- Option to not read last n number of barcodes
- Device can be a TCP/IP Client
- Configurable RS-232 settings
- Ability to configure GPIO Trigger delay
- Support for Fast Hardware Trigger (2.3 MP models only)
- Ability to set GPIOs to static high or low
- Added Job controls to clone, rename and delete jobs
- Support for Conditional Regular expressions for Barcode String Match
- Barcode String Match Regular expression support for ManyCode
- Backwards compatibility of Job files
- Decode and MV statistics in Aurora Focus and HMI stored and queried from device

V1.148.0001

RELEASE DATE: OCTOBER 2021

- Performance Enhancements
- Stability Improvements
- Implementation of licensing error messages in Web HMI
- Fixed issue of Job pausing on refresh of Web HMI
- Fixed issue of crash when clicking on "Communications" tab
- Fixed inability to "View" active Jobs from Job menu when Job utilizes a trigger that is automatic in nature.
- Fixed issue of slow decode rate when "Fast 1D/2D" license is enabled
- Added Configurable TCP/IP - results port and trigger string can be configured
- Support for GPIO > Hardware trigger (normal speed)
- Allow setting a job as Startup / Power up job
- Emulators now have license level
- Supports a total of 21 emulators
- ZJOB backwards compatibility and auto conversion of old job files (Windows only) - Seamless (live) tool changes when Upgrade licenses are applied

V1.134.0001 - Initial Product Release

RELEASE DATE: JULY 2021

- Bug Fixes
- Stability Improvements

Components

SW Component	Version
InstallShield	9.1.6
Aurora Focus™	9.1.6
Aurora Focus Deep Learning Editor	9.0.10631
BSP (xS40/70)	9.1.6 - CAAESS00-006-R08
BSP (FS42)	9.1.6 - CAAGGS00-006-R08
BSP (FS10/xS20)	9.1.6 - CAAFFS00-006-R08
BSP (FS80)	9.1.7 - SAAHMS00-001-R10
Aurora Focus™ HMI	10.0.7
Connectivity Gateway	1.2.10

The components are installed in the following folders:

Component	Location
Application	Program Files\Zebra Technologies\Aurora Focus/version
Job/Configuration Files	Users\Public\Documents\ZebraAurora/ModelType\Jobs
Saved Images	Users\Public\Documents\ZebraAurora/Captures/xSxxxxxx Camera

Minimum PC Requirements

Component	Location
CPU	Intel or AMD 64-bit processor
Random Access Memory Files	4 GB | Recommended: 8GB
GPU	GTX 1030, Intel UHD 620 or equivalent
Graphics Memory	2GB | Recommended: 4GB
Disk Space	4GB free disk space
Monitor Resolution	1280 x 800 | Recommended: 1920 x 1080
Operating System	Windows 10
Internet Connection	Required for software activation

Installation

Aurora Focus™ can be launched from the Windows 10 Start button.

- Start / All Programs / Zebra Scanner / Zebra Aurora Focus

Zebra Aurora Focus™ MV/FIS Configuration Utility

Description

Product Description Zebra Aurora Focus™ is a host-side application which allows for the configuration Zebra FS10, xS20, xS40, xS70, FS42, NS42 and FS80 devices. Zebra Aurora Focus™ Deep Learning Editor is companion software for training the Deep Learning Anomaly Detection tool. To use the editor, a physical or emulated NS42 device licensed for Anomaly Detection is required. It is installed independently and available in two versions: CPU and GPU versions. The GPU version is to train models using NVIDIA GPUs.

Device Compatibility

Compatibility Item	Details
Programming Modes	USB connected cable; M12 Ethernet

Supported Devices	Supported Device FW
FS40, VS40, FS70, VS70	**9.0.1000011 -** CAAESS00-006-R02
FS42, NS42	**9.0.1000011 -** CAAGGS00-006-R02
FS10, FS20, VS20	**9.0.1000011 -** CAAFFS00-006-R02
FS80	**9.0.2000013 -** SAAHMS00-001-R04

Release Notes

This document summarizes the following software release:

Software Version Number	Release Date
V9.0.12	December 2024

V9.0.12

RELEASE DATE: DECEMBER 2024

- Support for the NS42 Camera
- Support for the FS80 Camera. Note the following will be supported in future releases
    - JavaScript
    - FS80 Connectivity Gateway (Leader/Follower), GS20 is required
    - Image banks (Image Perfect)
    - Presentation mode
    - Communication with PLCs (Ethernet EIP, Modbus, Profinet)
    - ZETI command to control GPIO
    - SW license upgrade capability
    - Configuration barcodes
- Support for Multi-Function Integrated Light for FS80
- NEW tool: Deep Learning Anomaly Detection
- Deep Learning Editor (for training the Anomaly Detection tool)
- JavaScript Support
    - Output formatting
    - Output over TCP, Serial Port, CDC and basic HID (function keys not supported)
    - Editor with IntelliSense
    - Console logging for basic debugging
- Support for explicit serial trigger command
- Support for Multiple Aurora Focus versions installed on same host
- Option to select USB2 or USB3 for xS40/xS42/xS70
- Support for NPU-accelerated Deep Learning OCR
- Support for AutoTune on DPM with the Datacode tool
- Support for DPM QR code reading
- Support for reading QR Codes in Datacode tool
- Lower selectable heartbeat intervals
- Support for aggregating codes in presentation trigger mode
- Multiple Aurora Focus stability changes and enhancements
- Miscellaneous bug fixes
- Improved debugging for trigger overrun conditions when enabling logging on the camera

Known Issues

- Under some conditions, Datamatrix performance issues have been observed with Barcode tool.
  - Workaround: Use the Datacode tool.
- Under some conditions, barcode decode performance may be impacted.
  - Workarounds:
    - Disable unused symbologies.
    - Disable configuration barcodes (if not used) from the General menu.
    - Set expected module size appropriately.
- On occasion, dual update option on firmware update fails.
  - Workaround: Update firmware without using dual update. Then repeat the same firmware version update using "Force Update" with "Dual Update" checked.
- Sometimes HMI, after removing a license, does not update the screen.
  - Workaround: Refresh or switch tabs to update the screen.
- On some devices, USB port speed may not change from 2.0 to 3.0 when updated in Aurora.
  - Workaround:
    - Switch from 3.0 to 2.0 and apply settings.
    - Switch from 2.0 to 3.0 and apply settings.
    - Reboot the device.
- After very long idle durations, the JavaScript editor window may lose connection.
  - Workaround: Restart Aurora Focus.
- Autotune may fail if the center of image is black (no features) and the barcode is outside the center.
  - Workaround: Run autotune with the barcode closer to the center of field of view.
- FS80 currently does not support IP fallback to a default 169.254.x.x network IP.
  - Workaround: Use with a DHCP server.
- Reboot is required to see the Anomaly Detection tool in Aurora Focus after the device upgrade license is applied.
- Rarely, the job list in Aurora Focus does not show a job after reboot.
  - Workaround: Click Refresh at the top of the job list.
- Rarely, it may take a few seconds for the UI to update after selecting or deselecting all symbologies.
- When adding a very large number of tools to the VS flow builder, ROI boundaries may not be visible or performance issues may occur.
- When using Connectivity Gateway and configuring more than one GPIO, changes may not take effect immediately.
  - Workaround: Restart Runtime.
- WebHMI objects may not always load properly when opening the WebHMI page.
  - Workaround: Refresh the page.
- WebHMI objects may not always load properly on reboot.
  - Workaround: Refresh the page.
- If the device is factory reset over USB, or where NTP is not available, the device real-time clock may be reset.
  - Workaround: Open WebHMI and navigate to the license page to sync time with the browser host, or set the time manually.
- When saving images over FTP using a hostname, transfer delays may occur and the image is not transferred.
  - Workaround: Use the FTP server IP address instead of hostname.
- When using the Object Plus tools, operation may fail due to contrast value.
  - Workaround: Adjust contrast until it succeeds and the result is satisfactory.
- Rarely, OCR performance may be impacted depending on the inspected character set.
  - Workaround: Adjust ROI size.
- OCR performance is sometimes impacted when ROI is close to one or more characters.
  - Workaround: Increase ROI slightly.
- Some ZETI commands do not work with the shortened command format.
  - Workaround: Use the extended version of the ZETI command.
- In some cases, using many image banks causes slight processing delays.
  - Workaround: Reduce the number of image banks if possible.
- Device may not be discoverable in Aurora Focus when starting firmware update in HMI until update completes and the device reboots.
  - Workaround: Wait for firmware update completion.
- Firmware updates fail if memory consumption is already very high.
  - Workaround: Reboot the device before performing firmware update.
- Uninstalling InstallShield does not always remove all registry entries.
  - Workaround: Use `regedit` and manually delete the Aurora Focus folder under `Computer\HKEY_LOCAL_MACHINE\SOFTWARE\Zebra`.

V7.0.48000

RELEASE DATE: MAY 2024

- When editing a job for a camera, editing occurs directly on device and then timing values are more realistic.
- Support for the new FS42 camera
- Support for Separate (Explicit) TCP start and stop trigger strings
- Support for TCP control and outport on same port number.
- Connectivity Gateway: Support for backup and restore for configuration (HMI and ZETI)
- Connectivity Gateway: Support for level trigger Mode
- Connectivity Gateway: Separate start and stop TCP/Serial triggers
- Connectivity Gateway: Output at end of level trigger mode
- Connectivity Gateway: UI/UX Improvements
    - Now include helper tips in each section
    - Debugging tools added to bottom of Setup screen
- DL-OCR Match string settable by TCP command via ZETI
- Debug mode to save images in both level and continuous modes
- Support for saving images based on a time interval
- ZETI support for ASCII format when sending match string commands
- Multiple performance improvements
- Multiple Aurora Focus stability changes and enhancements
- Miscellaneous Bug Fixes

V6.2.70000

RELEASE DATE: FEBRUARY 2024

- Support for Liquid Lens focal distance using millimeters or inches.
- 2D Configuration barcode support to perform Autotune (Focus only)
- ZETI Command support to perform Autotune (Focus only)

V6.1.70000

RELEASE DATE: SEPTEMBER 2023

- Performance and Enhancements
    - Datacode Tool Improvements and Optimizations
    - Quantized Deep Learning OCR
    - FTP/SFTP folder storage suffix options
    - FTP/SFTP option to disable JSON results file
- ADF Output Enhancements
    - Support for Function Keys: F1-12
    - Support for Modifier Keys CTRL and ALT
- Option to Disable Tune and Trigger hardware buttons
- ZETI Interface improvements
- Backup and Restore feature improvements
- Miscellaneous Bug Fixes

V6.0.22000

RELEASE DATE: JULY 2023

- 2D Configuration Barcode Generation and Deployment
- Advanced Filtering and Formatting of Decode Data
- Multi-device management (Up to 5 devices at a time)
    - Firmware Update

    - Backup/Restore
    - Device Log Download
- Zebra Connectivity Gateway for Leader/Followers scanner configuration
- ZETI Command Interface Additional Command Support
- Aurora Focus UX Improvements
    - Improvements to Custom Formatting
    - Improved Localization
- Performance and Enhancements
    - Decode performance improvements.
    - Enhanced Session Based Triggering with Image Save
    - Additional Localization Enhancements

V5.0.13000

RELEASE DATE: FEBRUARY 2023

- Technology Previews
- Aurora Focus and HMI Language Support - Simplified Chinese, Korean, German, French and Spanish
- Expanded ZETI Interface Commands
- Tools
    - Deep Learning Based OCR for Fixed Industrial Scanners - Fully Released and Activated via Licenses
- Performance
    - Improvements to deployed job run consistency including Barcode tool timeout, decoding, and trigger timing consistency.
    - Level Trigger Enhancements
    - Triggered Image Buffering
- UX Improvements
    - Backup/Restore/Firmware Upgrade in Aurora Focus
- Device Configuration 1D Parameter Barcode Setup

V4.0.22000

RELEASE DATE: DECEMBER 2022

- Added Measure Object Tool
- Added Flaw Detection Edge and Intensity Tools
- Added Object Locate Counting
- Added Image Filters: Gradient, Gradient Horizontal, Vertical and Full
- MV Output Formatting on Supported Interfaces (Global or Interface Specific)
- Added Find and Replace to Barcode Decode Tool
- Improved 1D/2D Barcode Decode Performance
- Setting Barcode Match String via TCP or Serial input
- Added Image Download Capability via HMI
- Enhanced Session Based Triggering for Level, Continuous and Presentation Triggers w/Exit Criteria/Aggregation
- Technology Preview: Device Backup/Restore Functionality
- Performance and Stability Enhancements

V3.1.40000

RELEASE DATE: AUGUST 2022

- Added Reset on Trigger for Do Note Read Same Barcode Spanish HID Keyboard support
- Performance and Stability Enhancements

V3.0.10000

RELEASE DATE: JUNE 2022

- Added Edge Counting
- Added Deep-Learning Based OCR Tool (*in Beta*)
- Added Circle Find Tool
- Added Circle Diameter Measurement Tool
- Added a Circle ROI Type
- Added Open and Close Filters
- Added Erode Filter
- Added RS-232 Input Trigger
- Additional 2D Barcode Performance Improvements
- Added ADF to include Pass/Fail Data Types
- Added ADF to include Barcode Quality Metrics Data
- Performance and Stability Enhancements

V2.0.8000

RELEASE DATE: APRIL 2022

- Performance and Stability Enhancements

V2.0.7000

RELEASE DATE: MARCH 2022

- Performance and Stability Enhancements
- 1D Barcode Quality Metrics
- Added Object Locate Plus
- Added Blob Tools
- Added Binarize Filter/Tool
- Added Dilate Filter/Tool
- ImagePerfect(+) fixes & improvements
- Added Scale Calibration for Distance tool
- Added Advanced Pass/Fail Configuration (AND/OR logic for Job results) HID Keyboard Expansion - added French and German KB
- Added support for Codabar symbology
- Added Encoding type (Text/Base64) for ADF Full String data type and added option to show nonprintable characters as hex values
- For xS10/20, added external illumination support
- Added "File Path" option when using (S)FTP to Save images
- Added Ability to Open a Local Job (created with emulator) On a Real Connected Camera

V1.200.2001

RELEASE DATE: DECEMBER 2021

- Performance and Stability Enhancements
- Support for 5MP xS40 and xS70 models
- ImagePerfect(+) for FS models
- Option to output Barcode Location Co-ordinates
- Option to not read last n number of barcodes
- Device can be a TCP/IP Client
- Configurable RS-232 settings
- Ability to configure GPIO Trigger delay

- Support for Fast Hardware Trigger (2.3 MP models only)
- Ability to set GPIOs to static high or low
- Added Job controls to clone, rename and delete jobs
- Support for Conditional Regular expressions for Barcode String Match
- Barcode String Match Regular expression support for ManyCode
- Backwards compatibility of Job files
- Decode and MV statistics in Aurora Focus and HMI stored and queried from device

V1.148.0001

RELEASE DATE: OCTOBER 2021

- Performance Enhancements
- Stability Improvements
- Implementation of licensing error messages in Web HMI
- Fixed issue of Job pausing on refresh of Web HMI
- Fixed issue of crash when clicking on "Communications" tab Fixed inability to "View" active Jobs from Job menu when Job utilizes a trigger that is automatic in nature.
- Fixed issue of slow decode rate when "Fast 1D/2D" license is enabled
- Added Configurable TCP/IP - results port and trigger string can be configured
- Support for GPIO > Hardware trigger (normal speed)
- Allow setting a job as Startup / Power up job
- Emulators now have license level
- Supports a total of 21 emulators
- ZJOB backwards compatibility and auto conversion of old job files (Windows only) - Seamless (live) tool changes when Upgrade licenses are applied

V1.134.0001 - Initial Product Release

RELEASE DATE: JULY 2021

- Bug Fixes
- Stability Improvements

Components

SW Component	Version
InstallShield	9.0.12
Aurora Focus™	9.0.12
Aurora Focus Deep Learning Editor	9.0.10631
BSP (xS40/70)	**9.0.1000011 -** CAAESS00-002-R06
BSP (FS42)	**9.0.1000011 -** CAAGGS00-002-R06
BSP (FS10/xS20)	**9.0.1000011 -** CAAFFS00-002-R06
BSP (FS80)	**9.0.2000013 -** SAAHMS00-001-R04
Aurora Focus™ HMI	9.1.2
Connectivity Gateway	1.2.10

The components are installed in the following folders:

Component	Location
Application	Program Files\Zebra Technologies\Aurora Focus/version
Job/Configuration Files	Users\Public\Documents\ZebraAurora/ModelType\Jobs
Saved Images	Users\Public\Documents\ZebraAurora/Captures/xSxxxxxx Camera

Minimum PC Requirements

Component	Location
CPU	Intel or AMD 64-bit processor
Random Access Memory Files	4 GB | Recommended: 8GB
GPU	GTX 1030, Intel UHD 620 or equivalent
Graphics Memory	2GB | Recommended: 4GB
Disk Space	4GB free disk space
Monitor Resolution	1280 x 800 | Recommended: 1920 x 1080
Operating System	Windows 10
Internet Connection	Required for software activation

Installation

Aurora Focus™ can be launched from the Windows 10 Start button.

- Start / All Programs / Zebra Scanner / Zebra Aurora Focus

Zebra Aurora Focus™ MV/FIS Configuration Utility

Description

Product Description Zebra Aurora Focus™ is a host-side application which allows for the configuration Zebra FS10, xS20, xS40, xS70 and FS42 devices.

Device Compatibility

Compatibility Item	Details
Programming Modes	USB connected cable; M12 Ethernet

Supported Devices	Supported Device FW
FS40, VS40, FS70, VS70	CAAESS00-004-R09
FS42	CAAGGS00-004-R09
FS10, FS20, VS20	CAAFFS00-004-R09

Release Notes

This document summarizes the following software release:

Software Version Number	Release Date
V7.0.51000	October 2024

V7.0.51000

RELEASE DATE: OCTOBER 2024

- Updated to Support Hardware Models with the DPM 2D Standard Decoder License

Known Issues

- Under some conditions, barcode decode performance may be impacted.
  - Workarounds:
    - Disable unused symbologies.
    - Disable configuration barcodes (if not used).
    - Set expected module size appropriately.
- When using Connectivity Gateway and configuring to use more than one GPIO, changes may not take effect immediately.
  - Workaround: Restart Runtime.
- WebHMI objects may not always load properly when opening the WebHMI page.
  - Workaround: Refresh page.
- WebHMI objects may not always load properly on reboot.
  - Workaround: Refresh page.
- If factory resetting the device on a USB connection, or where NTP is not available, the device real-time clock may reset.
  - Workaround: Open WebHMI and navigate to the license page to sync time with the browser host, or set time manually.
- When saving images over FTP using a hostname, transfer delays may occur and the image is not transferred.
  - Workaround: Use FTP server IP address.
- When using Object Plus tools, operation may fail due to contrast value.
  - Workaround: Adjust contrast value until operation succeeds and results are satisfactory.
- Rarely, OCR performance may be impacted depending on character set.
  - Workaround: Adjust ROI size.
- OCR performance is sometimes impacted when ROI is close to one or more characters.
  - Workaround: Increase ROI slightly.
- Some ZETI commands do not work with the shortened command format.
  - Workaround: Use the extended ZETI command format.
- In some cases, using many image banks causes slight processing delays.
  - Workaround: Reduce number of image banks if possible.
- Device may not be discoverable in Aurora Focus when starting firmware update in HMI until update completes and device reboots.
  - Workaround: Wait for firmware update completion.

- Firmware updates fail if memory consumption is already very high.
    - Workaround: Reboot device before performing firmware update.
- Uninstalling InstallShield does not always remove all registry entries.
    - Workaround: Use `regedit` and manually delete the Aurora Focus folder under `Computer\HKEY_LOCAL_MACHINE\SOFTWARE\Zebra`.

V7.0.48000

RELEASE DATE: MAY 2024

- When editing a job for a camera, editing occurs directly on device and then timing values are more realistic.
- Support for the new FS42 camera
- Support for Separate (Explicit) TCP start and stop trigger strings - Support for TCP control and outport on same port number.
- Connectivity Gateway: Support for backup and restore for configuration (HMI and ZETI)
- Connectivity Gateway: Support for level trigger Mode
- Connectivity Gateway: Separate start and stop TCP/Serial triggers
- Connectivity Gateway: Output at end of level trigger mode
- Connectivity Gateway: UI/UX Improvements
    - Now include helper tips in each section
    - Debugging tools added to bottom of Setup screen - DL-OCR Match string settable by TCP command via ZETI
- Debug mode to save images in both level and continuous modes
- Support for saving images based on a time interval
- ZETI support for ASCII format when sending match string commands
- Multiple performance improvements
- Multiple Aurora Focus stability changes and enhancements
- Miscellaneous Bug Fixes

V6.2.70000

RELEASE DATE: FEBRUARY 2024

- Support for Liquid Lens focal distance using millimeters or inches.
- 2D Configuration barcode support to perform Autotune (Focus only)
- ZETI Command support to perform Autotune (Focus only)

V6.1.70000

RELEASE DATE: SEPTEMBER 2023 - Performance and Enhancements

- Datacode Tool Improvements and Optimizations
- Quantized Deep Learning OCR
- FTP/SFTP folder storage suffix options
- FTP/SFTP option to disable JSON results file
- ADF Output Enhancements
    - Support for Function Keys: F1-12
    - Support for Modifier Keys CTRL and ALT
- Option to Disable Tune and Trigger hardware buttons
- ZETI Interface improvements
- Backup and Restore feature improvements
- Miscellaneous Bug Fixes

V6.0.22000

RELEASE DATE: JULY 2023

- 2D Configuration Barcode Generation and Deployment
- Advanced Filtering and Formatting of Decode Data
- Multi-device management (Up to 5 devices at a time)
    - Firmware Update
    - Backup/Restore
    - Device Log Download
- Zebra Connectivity Gateway for Leader/Followers scanner configuration
- ZETI Command Interface Additional Command Support
- Aurora Focus UX Improvements
    - Improvements to Custom Formatting
    - Improved Localization
- Performance and Enhancements
    - Decode performance improvements.
    - Enhanced Session Based Triggering with Image Save
    - Additional Localization Enhancements

V5.0.13000

RELEASE DATE: FEBRUARY 2023

- Technology Previews
- Aurora Focus and HMI Language Support - Simplified Chinese, Korean, German, French and Spanish
- Expanded ZETI Interface Commands
- Tools
    - Deep Learning Based OCR for Fixed Industrial Scanners
- Fully Released - Activated via Licenses
- Performance
    - Improvements to deployed job run consistency including Barcode tool timeout, decoding, and trigger timing consistency.
    - Level Trigger Enhancements
    - Triggered Image Buffering
- UX Improvements
    - Backup/Restore/Firmware Upgrade in Aurora Focus
- Device Configuration 1D Parameter Barcode Setup

V4.0.22000

RELEASE DATE: DECEMBER 2022

- Added Measure Object Tool
- Added Flaw Detection Edge and Intensity Tools
- Added Object Locate Counting
- Added Image Filters: Gradient, Gradient Horizontal, Vertical and Full
- MV Output Formatting on Supported Interfaces (Global or Interface Specific)
- Added Find and Replace to Barcode Decode Tool
- Improved 1D/2D Barcode Decode Performance
- Setting Barcode Match String via TCP or Serial input
- Added Image Download Capability via HMI
- Enhanced Session Based Triggering for Level, Continuous and Presentation Triggers w/Exit Criteria/Aggregation
- Technology Preview: Device Backup/Restore Functionality
- Performance and Stability Enhancements

V3.1.40000

RELEASE DATE: AUGUST 2022

- Added Reset on Trigger for Do Note Read Same Barcode Spanish HID Keyboard support
- Performance and Stability Enhancements

V3.0.10000

RELEASE DATE: JUNE 2022

- Added Edge Counting
- Added Deep-Learning Based OCR Tool (in Beta)
- Added Circle Find Tool
- Added Circle Diameter Measurement Tool
- Added a Circle ROI Type
- Added Open and Close Filters
- Added Erode Filter
- Added RS-232 Input Trigger
- Additional 2D Barcode Performance Improvements
- Added ADF to include Pass/Fail Data Types
- Added ADF to include Barcode Quality Metrics Data
- Performance and Stability Enhancements

V2.0.8000

RELEASE DATE: APRIL 2022

- Performance and Stability Enhancements

V2.0.7000

RELEASE DATE: MARCH 2022

- Performance and Stability Enhancements
- 1D Barcode Quality Metrics
- Added Object Locate Plus
- Added Blob Tools
- Added Binarize Filter/Tool
- Added Dilate Filter/Tool
- ImagePerfect(+) fixes & improvements
- Added Scale Calibration for Distance tool
- Added Advanced Pass/Fail Configuration (AND/OR logic for Job results) HID Keyboard Expansion - added French and German KB
- Added support for Codabar symbology
- Added Encoding type (Text/Base64) for ADF Full String data type and added option to show nonprintable characters as hex values
- For xS10/20, added external illumination support
- Added "File Path" option when using (S)FTP to Save images
- Added Ability to Open a Local Job (created with emulator) On a Real Connected Camera

V1.200.2001

RELEASE DATE: DECEMBER 2021

- Performance and Stability Enhancements
- Support for 5MP xS40 and xS70 models
- ImagePerfect(+) for FS models
- Option to output Barcode Location Co-ordinates
- Option to not read last n number of barcodes
- Device can be a TCP/IP Client
- Configurable RS-232 settings
- Ability to configure GPIO Trigger delay
- Support for Fast Hardware Trigger (2.3 MP models only)
- Ability to set GPIOs to static high or low
- Added Job controls to clone, rename and delete jobs
- Support for Conditional Regular expressions for Barcode String Match
- Barcode String Match Regular expression support for ManyCode
- Backwards compatibility of Job files
- Decode and MV statistics in Aurora Focus and HMI stored and queried from device

V1.148.0001

RELEASE DATE: OCTOBER 2021

- Performance Enhancements
- Stability Improvements
- Implementation of licensing error messages in Web HMI
- Fixed issue of Job pausing on refresh of Web HMI
- Fixed issue of crash when clicking on "Communications" tab Fixed inability to "View" active Jobs from Job menu when Job utilizes a trigger that is automatic in nature.
- Fixed issue of slow decode rate when "Fast 1D/2D" license is enabled
- Added Configurable TCP/IP - results port and trigger string can be configured
- Support for GPIO > Hardware trigger (normal speed)
- Allow setting a job as Startup / Power up job
- Emulators now have license level
- Supports a total of 21 emulators
- ZJOB backwards compatibility and auto conversion of old job files (Windows only) - Seamless (live) tool changes when Upgrade licenses are applied

V1.134.0001 - Initial Product Release

RELEASE DATE: JULY 2021

- Bug Fixes
- Stability Improvements

Components

SW Component	Version
InstallShield	7.0.51000
Aurora Focus™	7.0.51
BSP (xS40/70)	CAAESS00-004-R09
BSP (FS42)	CAAGGS00-004-R09
BSP (FS10/xS20)	CAAFFS00-004-R09
Aurora Focus™ HMI	4.5.7
Connectivity Gateway	1.2.7

The components are installed in the following folders:

Component	Location
Application	Program Files\Zebra Technologies\Aurora
Job/Configuration Files	Users\Public\Documents\ZebraAurora\ModelType\Jobs
Saved Images	Users\Public\Documents\ZebraAurora\Captures\xSxxxxxx Camera

Minimum PC Requirements

Component	Location
CPU	Intel or AMD 64-bit processor
Random Access Memory Files	4 GB | Recommended: 8GB
GPU	GTX 1030, Intel UHD 620 or equivalent
Graphics Memory	2GB | Recommended: 4GB
Disk Space	4GB free disk space
Monitor Resolution	1280 x 800 | Recommended: 1920 x 1080
Operating System	Windows 10
Internet Connection	Required for software activation

Installation

Aurora Focus™ can be launched from the Windows 10 Start button.

- Start / All Programs / Zebra Scanner / Zebra Aurora Focus Page

Zebra Aurora Focus™ MV/FIS Configuration Utility

Description

Product Description Zebra Aurora Focus™ is a host-side application which allows for the configuration Zebra FS10, xS20, xS40, xS70 and FS42 devices.

Device Compatibility

Compatibility Item	Details
Programming Modes	USB connected cable; M12 Ethernet

Supported Devices	Supported Device FW
FS40, VS40, FS70, VS70	CAAESS00-004-R08
FS42	CAAGGS00-004-R08
FS10, FS20, VS20	CAAFFS00-004-R08

Release Notes

This document summarizes the following software release:

Software Version Number	Release Date
V7.0.48000	May 2024

V7.0.48000

RELEASE DATE: MAY 2024

- When editing a job for a camera, editing occurs directly on device and then timing values are more realistic.
- Support for the new FS42 camera
- Support for TCP control and outport on same port number.
- Connectivity Gateway: Supports Backup and Restore for Configuration
- Connectivity Gateway: Output at end of Level trigger mode
- Connectivity Gateway: UI/UX Improvements
- DL-OCR Match String settable by TCP command
- When debug mode enabled in Aurora Focus, support for saving images based on time interval
- ZETI support for ASCII format when sending match string commands
- Multiple Aurora Focus stability changes and enchantements

Known Issues

- Under some conditions, barcode decode performance may be impacted.
  - Resolutions:
    - Disable unused symbologies.
    - Disable configuration barcodes (if not used).
    - Set expected module size appropriately.
- When using Connectivity Gateway and configuring to use more than one GPIO, changes may not take effect immediately.
  - Workaround: Restart Runtime.
- WebHMI objects may not always load properly when opening the WebHMI page.
  - Workaround: Refresh page.
- WebHMI objects may not always load properly on reboot.
  - Workaround: Refresh page.
- If Factory Resetting the device on a USB connection, or where NTP is not available, the device real time clock may be reset.
  - Workaround: Open WebHMI and navigate to the license page to sync time with the browser host or set it manually.
- When saving images over FTP using a hostname, delays in transfer sometimes occur and the image is not transferred.
  - Workaround: Use FTP server IP address.
- When using the Object Plus tools, it will sometimes fail due to the contrast value.
  - Workaround: Adjust contrast value until it succeeds and results are satisfactory.
- Rarely, OCR performance can be impacted depending on the character set being inspected.

    - Workaround: Adjust ROI size.
- OCR performance is sometimes impacted when the ROI (Region of Interest) is close to one or more of the characters.
    - Workaround: Increase ROI slightly.
- Some ZETI commands will not work with the command's shortened format.
    - Workaround: Use the extended version of the ZETI command.
- In some cases, using many image-banks causes slight delays in processing.
    - Workaround: If possible, reduce number of image banks.
- Device will sometimes not be discoverable in Aurora Focus when starting firmware update in HMI until after update is complete and device reboots.
    - Workaround: Wait until firmware update completes.
- Firmware updates will fail if memory consumption is already exceedingly high.
    - Workaround: Reboot the device before performing firmware update.
- Uninstalling the InstallShield does not always remove all registry entries.
    - Workaround: Use `regedit` and manually delete the Aurora Focus folder under `Computer\HKEY_LOCAL_MACHINE\SOFTWARE\Zebra`.

V6.2.70000

RELEASE DATE: FEBRUARY 2024

- Support for Liquid Lens focal distance using millimeters or inches.
- 2D Configuration barcode support to perform Autotune (Focus only)
- ZETI Command support to perform Autotune (Focus only)

V6.1.70000

RELEASE DATE: SEPTEMBER 2023

- Performance and Enhancements
    - Datacode Tool Improvements and Optimizations
    - Quantized Deep Learning OCR
    - FTP/SFTP folder storage suffix options
    - FTP/SFTP option to disable JSON results file
- ADF Output Enhancements
    - Support for Function Keys: F1-12
    - Support for Modifier Keys CTRL and ALT
- Option to Disable Tune and Trigger hardware buttons
- ZETI Interface improvements
- Backup and Restore feature improvements
- Miscellaneous Bug Fixes

V6.0.22000

RELEASE DATE: JULY 2023

- 2D Configuration Barcode Generation and Deployment
- Advanced Filtering and Formatting of Decode Data
- Multi-device management (Up to 5 devices at a time)
    - Firmware Update
    - Backup/Restore
    - Device Log Download
- Zebra Connectivity Gateway for Leader/Followers scanner configuration
- ZETI Command Interface Additional Command Support
- Aurora Focus UX Improvements
    - Improvements to Custom Formatting
    - Improved Localization
- Performance and Enhancements

    - Decode performance improvements.
    - Enhanced Session Based Triggering with Image Save
    - Additional Localization Enhancements

V5.0.13000

RELEASE DATE: FEBRUARY 2023

- Technology Previews
    - Aurora Focus and HMI Language Support - Simplified Chinese, Korean, German, French and Spanish
    - Expanded ZETI Interface Commands
- Tools
    - Deep Learning Based OCR for Fixed Industrial Scanners - Fully Released and Activated via Licenses
- Performance
    - Improvements to deployed job run consistency including Barcode tool timeout, decoding, and trigger timing consistency.
    - Level Trigger Enhancements
    - Triggered Image Buffering
- UX Improvements
    - Backup/Restore/Firmware Upgrade in Aurora Focus
- Device Configuration 1D Parameter Barcode Setup

V4.0.22000

RELEASE DATE: DECEMBER 2022

- Added Measure Object Tool
- Added Flaw Detection Edge and Intensity Tools
- Added Object Locate Counting
- Added Image Filters: Gradient, Gradient Horizontal, Vertical and Full
- MV Output Formatting on Supported Interfaces (Global or Interface Specific)
- Added Find and Replace to Barcode Decode Tool
- Improved 1D/2D Barcode Decode Performance
- Setting Barcode Match String via TCP or Serial input
- Added Image Download Capability via HMI
- Enhanced Session Based Triggering for Level, Continuous and Presentation Triggers w/Exit Criteria/Aggregation
- Technology Preview: Device Backup/Restore Functionality
- Performance and Stability Enhancements

V3.1.40000

RELEASE DATE: AUGUST 2022

- Added Reset on Trigger for Do Note Read Same Barcode
- Spanish HID Keyboard support
- Performance and Stability Enhancements

V3.0.10000

RELEASE DATE: JUNE 2022

- Added Edge Counting
- Added Deep-Learning Based OCR Tool (*in Beta*)
- Added Circle Find Tool
- Added Circle Diameter Measurement Tool
- Added a Circle ROI Type

- Added Open and Close Filters
- Added Erode Filter
- Added RS-232 Input Trigger
- Additional 2D Barcode Performance Improvements
- Added ADF to include Pass/Fail Data Types
- Added ADF to include Barcode Quality Metrics Data
- Performance and Stability Enhancements

V2.0.8000

RELEASE DATE: APRIL 2022

- Performance and Stability Enhancements

V2.0.7000

RELEASE DATE: MARCH 2022

- Performance and Stability Enhancements
- 1D Barcode Quality Metrics
- Added Object Locate Plus
- Added Blob Tools
- Added Binarize Filter/Tool
- Added Dilate Filter/Tool
- ImagePerfect(+) fixes & improvements
- Added Scale Calibration for Distance tool
- Added Advanced Pass/Fail Configuration (AND/OR logic for Job results)
- HID Keyboard Expansion - added French and German KB
- Added support for Codabar symbology
- Added Encoding type (Text/Base64) for ADF Full String data type and added option to show nonprintable characters as hex values
- For xS10/20, added external illumination support
- Added "File Path" option when using (S)FTP to Save images
- Added Ability to Open a Local Job (created with emulator) On a Real Connected Camera

V1.200.2001

RELEASE DATE: DECEMBER 2021

- Performance and Stability Enhancements
- Support for 5MP xS40 and xS70 models
- ImagePerfect(+) for FS models
- Option to output Barcode Location Co-ordinates
- Option to not read last n number of barcodes
- Device can be a TCP/IP Client
- Configurable RS-232 settings
- Ability to configure GPIO Trigger delay
- Support for Fast Hardware Trigger (2.3 MP models only)
- Ability to set GPIOs to static high or low
- Added Job controls to clone, rename and delete jobs
- Support for Conditional Regular expressions for Barcode String Match
- Barcode String Match Regular expression support for ManyCode
- Backwards compatibility of Job files
- Decode and MV statistics in Aurora Focus and HMI stored and queried from device

V1.148.0001

RELEASE DATE: OCTOBER 2021

- Performance Enhancements
- Stability Improvements
- Implementation of licensing error messages in Web HMI
- Fixed issue of Job pausing on refresh of Web HMI
- Fixed issue of crash when clicking on "Communications" tab
- Fixed inability to "View" active Jobs from Job menu when Job utilizes a trigger that is automatic in nature.
- Fixed issue of slow decode rate when "Fast 1D/2D" license is enabled
- Added Configurable TCP/IP - results port and trigger string can be configured
- Support for GPIO > Hardware trigger (normal speed)
- Allow setting a job as Startup / Power up job
- Emulators now have license level
- Supports a total of 21 emulators
- ZJOB backwards compatibility and auto conversion of old job files (Windows only) - Seamless (live) tool changes when Upgrade licenses are applied

V1.134.0001 - Initial Product Release

RELEASE DATE: JULY 2021

- Bug Fixes
- Stability Improvements

Components

SW Component	Version
InstallShield	7.0.48000
Aurora Focus™	7.0.48
BSP (xS40/70)	CAAESS00-004-R08
BSP (FS42)	CAAGGS00-004-R08
BSP (FS10/xS20)	CAAFFS00-004-R08
Aurora Focus™ HMI	4.5.7
Connectivity Gateway	1.2.7

The components are installed in the following folders:

Component	Location
Application	Program Files\Zebra Technologies\Aurora
Job/Configuration Files	Users\Public\Documents\ZebraAurora/ModelType\Jobs
Saved Images	Users\Public\Documents\ZebraAurora/Captures/xSxxxxxx Camera

Minimum PC Requirements

Component	Location
CPU	Intel or AMD 64-bit processor
Random Access Memory Files	4 GB | Recommended: 8GB
GPU	GTX 1030, Intel UHD 620 or equivalent
Graphics Memory	2GB | Recommended: 4GB
Disk Space	4GB free disk space
Monitor Resolution	1280 x 800 | Recommended: 1920 x 1080
Operating System	Windows 10
Internet Connection	Required for software activation

Installation

Aurora Focus™ can be launched from the Windows 10 Start button.

- Start / All Programs / Zebra Scanner / Zebra Aurora Focus Page

Zebra Aurora Focus™ MV/FIS Configuration Utility

Description

Product Description Zebra Aurora Focus™ is a host-side application which allows for the configuration Zebra FS10, xS20, xS40 and xS70 devices.

Device Compatibility

Compatibility Item	Details
Programming Modes	USB connected cable; M12 Ethernet

Supported Devices	Supported Device FW
FS40, VS40, FS70, VS70	CAAESS00-003-R25
FS10, FS20, VS20	CAAFFS00-003-R25

Release Notes

This document summarizes the following software release:

Software Version Number	Release Date
V6.2.70000	February 2024

V6.2.70000

Workaround: use `regedit` and manually delete the Aurora Focus folder under `Computer\HKEY_LOCAL_MACHINE\SOFTWARE\Zebra`.

- Support for Liquid Lens focal distance using millimeters or inches.
- Tools
  - Deep Learning Based OCR for Fixed Industrial Scanners - Fully Released and Activated via Licenses
- ZETI Command support to perform Autotune (Focus only)

Known Issues

- OCR performance is sometimes impacted when the ROI (Region of Interest) is close to one or more of the characters.
  - Workaround: Increase ROI slightly.
- When using custom Output Formatting and enabling different interfaces (i.e., TCP and Serial), and dragging the result field tag, it will sometimes appear in the other interface's output formatting.
  - Workaround: Click on the tag instead of dragging it.
- When connected to Aurora through USB, after long durations, rarely the device may no longer be seen.
  - Workaround: Unplug and re-plug the USB cable.
- Using "index" for the filename suffix when saving images to the device is currently not supported.
  - Workaround: Used ΓÇÿDateTime' suffix
- When Tune and TRIG buttons are disabled, re-enabling after reboot is not supported.
  - Workaround: Reboot the device for the changes to take effect.
- If Factory Resetting the device on a USB connection, or where NTP is not available, the device real time clock may be reset.
  - Workaround: Open up the WebHMI screen and navigate to the license page to sync the time with the browser host or set the time manually.
- When saving images over FTP using a hostname, delays in transfer sometimes occur and the image is not transferred.
  - Workaround: Use IP address of FTP server instead.
- When using the Object Plus tools, it will sometimes fail due to the contrast value.

    - Workaround: Adjust contrast value until it succeeds, and the result is satisfactory.
- Rarely, OCR performance can be impacted depending on the character set being inspected.
    - Workaround: Adjustment of the size of the ROI can sometimes improve performance
- Some ZETI commands will not work with the command's shortened format.
    - Workaround: Use the extended version of the ZETI command
- When configuring a leader/follower barcode setup, the serial port can sometimes conflict with a job on the device.
    - Workaround: Disable the Serial port in Aurora Focus.
- Flaw detection tools must click ΓÇÿUndo' twice to perform the undo operation.
    - Workaround: Click undo twice
- In some cases, using many image-banks causes slight delays in processing.
    - Workaround: If possible, reduce number of image banks
- Device will sometimes not be discoverable in Aurora Focus when starting firmware update in HMI until after update is complete and device reboots.
    - Workaround: Wait until firmware update completes.
- Firmware updates will fail if memory consumption is already exceedingly high.
    - Workaround: Reboot the device before performing firmware update.
- Uninstalling the InstallShield does not always remove all registry entries.
    - Workaround: use `regedit` and manually delete the Aurora Focus folder under `Computer\HKEY_LOCAL_MACHINE\SOFTWARE\Zebra`.

V6.1.70000

RELEASE DATE: SEPTEMBER 2023

- Performance and Enhancements
    - Datacode Tool Improvements and Optimizations
    - Quantized Deep Learning OCR
    - FTP/SFTP folder storage suffix options
    - FTP/SFTP option to disable JSON results file
- ADF Output Enhancements
    - Support for Function Keys: F1-12
    - Support for Modifier Keys CTRL and ALT
- Option to Disable Tune and Trigger hardware buttons
- ZETI Interface improvements
- Backup and Restore feature improvements
- Miscellaneous Bug Fixes

V6.0.22000

RELEASE DATE: JULY 2023

- 2D Configuration Barcode Generation and Deployment
- Advanced Filtering and Formatting of Decode Data
- Multi-device management (Up to 5 devices at a time)
    - Firmware Update
    - Backup/Restore
    - Device Log Download
- Zebra Connectivity Gateway for Leader/Followers scanner configuration
- ZETI Command Interface Additional Command Support
- Aurora Focus UX Improvements
    - Improvements to Custom Formatting
    - Improved Localization
- Performance and Enhancements
    - Decode performance improvements.
    - Enhanced Session Based Triggering with Image Save
    - Additional Localization Enhancements

V5.0.13000

RELEASE DATE: FEBRUARY 2023

- Technology Previews
  - Aurora Focus and HMI Language Support - Simplified Chinese, Korean, German, French and Spanish
  - Expanded ZETI Interface Commands
- Tools
  - Deep Learning Based OCR for Fixed Industrial Scanners - Fully Released and Activated via Licenses
- Performance
  - Improvements to deployed job run consistency including Barcode tool timeout, decoding, and trigger timing consistency.
  - Level Trigger Enhancements
  - Triggered Image Buffering
- UX Improvements
  - Backup/Restore/Firmware Upgrade in Aurora Focus
- Device Configuration 1D Parameter Barcode Setup

V4.0.22000

RELEASE DATE: DECEMBER 2022

- Added Measure Object Tool
- Added Flaw Detection Edge and Intensity Tools
- Added Object Locate Counting
- Added Image Filters: Gradient, Gradient Horizontal, Vertical and Full
- MV Output Formatting on Supported Interfaces (Global or Interface Specific)
- Added Find and Replace to Barcode Decode Tool
- Improved 1D/2D Barcode Decode Performance
- Setting Barcode Match String via TCP or Serial input
- Added Image Download Capability via HMI
- Enhanced Session Based Triggering for Level, Continuous and Presentation Triggers w/Exit Criteria/Aggregation
- Technology Preview: Device Backup/Restore Functionality
- Performance and Stability Enhancements

V3.1.40000

RELEASE DATE: AUGUST 2022

- Added Reset on Trigger for Do Note Read Same Barcode
- Spanish HID Keyboard support
- Performance and Stability Enhancements

V3.0.10000

RELEASE DATE: JUNE 2022

- Added Edge Counting
- Added Deep-Learning Based OCR Tool (*in Beta*)
- Added Circle Find Tool
- Added Circle Diameter Measurement Tool
- Added a Circle ROI Type
- Added Open and Close Filters
- Added Erode Filter
- Added RS-232 Input Trigger
- Additional 2D Barcode Performance Improvements

- Added ADF to include Pass/Fail Data Types
- Added ADF to include Barcode Quality Metrics Data
- Performance and Stability Enhancements

V2.0.8000

RELEASE DATE: APRIL 2022

- Performance and Stability Enhancements

V2.0.7000

RELEASE DATE: MARCH 2022

- Performance and Stability Enhancements
- 1D Barcode Quality Metrics
- Added Object Locate Plus
- Added Blob Tools
- Added Binarize Filter/Tool
- Added Dilate Filter/Tool
- ImagePerfect(+) fixes & improvements
- Added Scale Calibration for Distance tool
- Added Advanced Pass/Fail Configuration (AND/OR logic for Job results)
- HID Keyboard Expansion - added French and German KB
- Added support for Codabar symbology
- Added Encoding type (Text/Base64) for ADF Full String data type and added option to show nonprintable characters as hex values
- For xS10/20, added external illumination support
- Added "File Path" option when using (S)FTP to Save images
- Added Ability to Open a Local Job (created with emulator) On a Real Connected Camera

V1.200.2001

RELEASE DATE: DECEMBER 2021

- Performance and Stability Enhancements
- Support for 5MP xS40 and xS70 models
- ImagePerfect(+) for FS models
- Option to output Barcode Location Co-ordinates
- Option to not read last n number of barcodes
- Device can be a TCP/IP Client
- Configurable RS-232 settings
- Ability to configure GPIO Trigger delay
- Support for Fast Hardware Trigger (2.3 MP models only)
- Ability to set GPIOs to static high or low
- Added Job controls to clone, rename and delete jobs
- Support for Conditional Regular expressions for Barcode String Match
- Barcode String Match Regular expression support for ManyCode
- Backwards compatibility of Job files
- Decode and MV statistics in Aurora Focus and HMI stored and queried from device

V1.148.0001

RELEASE DATE: OCTOBER 2021

- Performance Enhancements

- Stability Improvements
- Implementation of licensing error messages in Web HMI
- Fixed issue of Job pausing on refresh of Web HMI
- Fixed issue of crash when clicking on "Communications" tab
- Fixed inability to "View" active Jobs from Job menu when Job utilizes a trigger that is automatic in nature.
- Fixed issue of slow decode rate when "Fast 1D/2D" license is enabled
- Added Configurable TCP/IP - results port and trigger string can be configured
- Support for GPIO > Hardware trigger (normal speed)
- Allow setting a job as Startup / Power up job
- Emulators now have license level
- Supports a total of 21 emulators
- ZJOB backwards compatibility and auto conversion of old job files (Windows only) - Seamless (live) tool changes when Upgrade licenses are applied

V1.134.0001 - Initial Product Release

RELEASE DATE: JULY 2021

- Bug Fixes
- Stability Improvements

Components

SW Component	Version
InstallShield	6.2.70000
Aurora Focus™	6.2.7
BSP (xS40/70)	CAAESS00-003-R25
BSP (FS10/xS20)	CAAFFS00-003-R25
Aurora Focus™ HMI	2.4.2
Connectivity Gateway	1.1.15

The components are installed in the following folders:

Component	Location
Application	Program Files\Zebra Technologies\Zebra Aurora
Job/Configuration Files	Users\Public\Documents\ZebraAurora/ModelType\Jobs
Saved Images	Users\Public\Documents\ZebraAurora/Captures/xSxxxxxx Camera

Minimum PC Requirements

Component	Location
CPU	Intel or AMD 64-bit processor
Random Access Memory Files	4 GB | Recommended: 8GB
GPU	GTX 1030, Intel UHD 620 or equivalent
Graphics Memory	2GB | Recommended: 4GB
Disk Space	4GB free disk space
Monitor Resolution	1280 x 800 | Recommended: 1920 x 1080
Operating System	Windows 10
Internet Connection	Required for software activation

Installation

Aurora Focus™ can be launched from the Windows 10 Start button.

- Start / All Programs / Zebra Scanner / Zebra Aurora Focus

Zebra Aurora Focus™ MV/FIS Configuration Utility

Description

Product Description Zebra Aurora Focus™ is a host-side application which allows for the configuration Zebra FS10, xS20, xS40 and xS70 devices.

Device Compatibility

Compatibility Item	Details
Programming Modes	USB connected cable; M12 Ethernet

Supported Devices	Supported Device FW
FS40, VS40, FS70, VS70	CAAESS00-003-R18
FS10, FS20, VS20	CAAFFS00-003-R18

Release Notes

This document summarizes the following software release:

Software Version Number	Release Date
V6.1.70000	September 2023

V6.1.70000

RELEASE DATE: SEPTEMBER 2023

- Performance and Enhancements
  - Datacode Tool Improvements and Optimizations
- Quantized Deep Learning OCR
- FTP/SFTP folder storage suffix options
  - FTP/SFTP option to disable JSON results file
- ADF Output Enhancements
  - Support for Function Keys: F1-12
  - Support for Modifier Keys CTRL and ALT
- Option to Disable Tune and Trigger hardware buttons
- ZETI Interface improvements
- Backup and Restore feature improvements
- Miscellaneous Bug Fixes

Known Issues

- OCR performance is sometimes impacted when the ROI (Region of Interest) is close to one or more of the characters.
  - Workaround: Increase ROI slightly.
- When using custom Output Formatting and enabling different interfaces (i.e., TCP and Serial), and dragging the result field tag, it will sometimes appear in the other interface's output formatting.
  - Workaround: Click on the tag instead of dragging it.
- When connected to Aurora through USB, after long durations, rarely the device may no longer be seen.
  - Workaround: Unplug and re-plug the USB cable.
- Using "index" for the filename suffix when saving images to the device is currently not supported.
  - Workaround: Used ΓÇÿDateTime' suffix
- When Tune and TRIG buttons are disabled, re-enabling after reboot is not supported.
  - Workaround: Reboot the device for the changes to take effect.
- If Factory Resetting the device on a USB connection, or where NTP is not available, the device real time clock may be reset.
  - Workaround: Open up the WebHMI screen and navigate to the license page to sync the time with the browser host or set the time manually.
- When saving images over FTP using a hostname, delays in transfer sometimes occur and the image is not transferred.
  - Workaround: Use IP address of FTP server instead.
- When using the Object Plus tools, it will sometimes fail due to the contrast value.
  - Workaround: Adjust contrast value until it succeeds, and the result is satisfactory.
- Rarely, OCR performance can be impacted depending on the character set being inspected.

    - Workaround: Adjustment of the size of the ROI can sometimes improve performance
- Some ZETI commands will not work with the command's shortened format.
    - Workaround: Use the extended version of the ZETI command
- When configuring a leader/follower barcode setup, the serial port can sometimes conflict with a job on the device.
    - Workaround: Disable the Serial port in Aurora Focus.
- Flaw detection tools must click ΓÇÿUndo' twice to perform the undo operation.
    - Workaround: Click undo twice
- In some cases, using many image-banks causes slight delays in processing.
    - Workaround: If possible, reduce number of image banks
- Device will sometimes not be discoverable in Aurora Focus when starting firmware update in HMI until after update is complete and device reboots.
    - Workaround: Wait until firmware update completes.
- Firmware updates will fail if memory consumption is already exceedingly high.
    - Workaround: Reboot the device before performing firmware update.
- Uninstalling the InstallShield does not always remove all registry entries.
    - Workaround: use `regedit` and manually delete the Aurora Focus folder under `Computer\HKEY_LOCAL_MACHINE\SOFTWARE\Zebra`.

V6.0.22000

RELEASE DATE: JULY 2023

- 2D Configuration Barcode Generation and Deployment
- Advanced Filtering and Formatting of Decode Data
- Multi-device management (Up to 5 devices at a time)
    - Firmware Update
    - Backup/Restore
    - Device Log Download
- Zebra Connectivity Gateway for Leader/Followers scanner configuration
- ZETI Command Interface Additional Command Support
- Aurora Focus UX Improvements
    - Improvements to Custom Formatting
    - Improved Localization
- Performance and Enhancements
    - Decode performance improvements.
    - Enhanced Session Based Triggering with Image Save
    - Additional Localization Enhancements

V5.0.13000

RELEASE DATE: FEBRUARY 2023

- Technology Previews
    - Aurora Focus and HMI Language Support - Simplified Chinese, Korean, German, French and Spanish
    - Expanded ZETI Interface Commands
- Tools
    - Deep Learning Based OCR for Fixed Industrial Scanners - Fully Released and Activated via Licenses
- Performance
    - Improvements to deployed job run consistency including Barcode tool timeout, decoding, and trigger timing consistency.
    - Level Trigger Enhancements
    - Triggered Image Buffering UX Improvements
    - Backup/Restore/Firmware Upgrade in Aurora Focus
- Device Configuration 1D Parameter Barcode Setup

V4.0.22000

RELEASE DATE: DECEMBER 2022

- Added Measure Object Tool
- Added Flaw Detection Edge and Intensity Tools
- Added Object Locate Counting
- Added Image Filters: Gradient, Gradient Horizontal, Vertical and Full
- MV Output Formatting on Supported Interfaces (Global or Interface Specific)
- Added Find and Replace to Barcode Decode Tool
- Improved 1D/2D Barcode Decode Performance
- Setting Barcode Match String via TCP or Serial input
- Added Image Download Capability via HMI
- Enhanced Session Based Triggering for Level, Continuous and Presentation Triggers w/Exit Criteria/Aggregation
- Technology Preview: Device Backup/Restore Functionality
- Performance and Stability Enhancements

V3.1.40000

RELEASE DATE: AUGUST 2022

- Added Reset on Trigger for Do Note Read Same Barcode
- Spanish HID Keyboard support
- Performance and Stability Enhancements

V3.0.10000

RELEASE DATE: JUNE 2022

- Added Edge Counting
- Added Deep-Learning Based OCR Tool (*in Beta*)
- Added Circle Find Tool
- Added Circle Diameter Measurement Tool
- Added a Circle ROI Type Added Open and Close Filters
- Added Erode Filter
- Added RS-232 Input Trigger
- Additional 2D Barcode Performance Improvements
- Added ADF to include Pass/Fail Data Types
- Added ADF to include Barcode Quality Metrics Data
- Performance and Stability Enhancements

V2.0.8000

RELEASE DATE: APRIL 2022

- Performance and Stability Enhancements

V2.0.7000

RELEASE DATE: MARCH 2022

- Performance and Stability Enhancements
- 1D Barcode Quality Metrics
- Added Object Locate Plus

- Added Blob Tools
- Added Binarize Filter/Tool
- Added Dilate Filter/Tool
- ImagePerfect(+) fixes & improvements
- Added Scale Calibration for Distance tool
- Added Advanced Pass/Fail Configuration (AND/OR logic for Job results)
- HID Keyboard Expansion - added French and German KB
- Added support for Codabar symbology
- Added Encoding type (Text/Base64) for ADF Full String data type and added option to show nonprintable characters as hex values
- For xS10/20, added external illumination support
- Added "File Path" option when using (S)FTP to Save images
- Added Ability to Open a Local Job (created with emulator) On a Real Connected Camera

V1.200.2001

RELEASE DATE: DECEMBER 2021

Performance and Stability Enhancements

- Support for 5MP xS40 and xS70 models
- ImagePerfect(+) for FS models
- Option to output Barcode Location Co-ordinates
- Option to not read last n number of barcodes
- Device can be a TCP/IP Client
- Configurable RS-232 settings
- Ability to configure GPIO Trigger delay
- Support for Fast Hardware Trigger (2.3 MP models only)
- Ability to set GPIOs to static high or low
- Added Job controls to clone, rename and delete jobs
- Support for Conditional Regular expressions for Barcode String Match
- Barcode String Match Regular expression support for ManyCode
- Backwards compatibility of Job files
- Decode and MV statistics in Aurora Focus and HMI stored and queried from device

V1.148.0001

RELEASE DATE: OCTOBER 2021

- Performance Enhancements
- Stability Improvements
- Implementation of licensing error messages in Web HMI
- Fixed issue of Job pausing on refresh of Web HMI
- Fixed issue of crash when clicking on "Communications" tab
- Fixed inability to "View" active Jobs from Job menu when Job utilizes a trigger that is automatic in nature.
- Fixed issue of slow decode rate when "Fast 1D/2D" license is enabled
- Added Configurable TCP/IP - results port and trigger string can be configured
- Support for GPIO > Hardware trigger (normal speed)
- Allow setting a job as Startup / Power up job
- Emulators now have license level
- Supports a total of 21 emulators
- ZJOB backwards compatibility and auto conversion of old job files (Windows only) - Seamless (live) tool changes when Upgrade licenses are applied

V1.134.0001 - Initial Product Release

RELEASE DATE: JULY 2021

- Bug Fixes
- Stability Improvements

Components

SW Component	Version
InstallShield	6.1.70000
Aurora Focus™	6.1.7
BSP (xS40/70)	CAAESS00-003-R18
BSP (FS10/xS20)	CAAFFS00-003-R18
Aurora Focus™ HMI	2.4.2
Connectivity Gateway	1.1.15

The components are installed in the following folders:

Component	Location
Application	Program Files\Zebra Technologies\Zebra Aurora
Job/Configuration Files	Users\Public\Documents\ZebraAurora/ModelType\Jobs
Saved Images	Users\Public\Documents\ZebraAurora/Captures/xSxxxxxx Camera

Minimum PC Requirements

Component	Location
CPU	Intel or AMD 64-bit processor
Random Access Memory Files	4 GB | Recommended: 8GB
GPU	GTX 1030, Intel UHD 620 or equivalent
Graphics Memory	2GB | Recommended: 4GB
Disk Space	4GB free disk space
Monitor Resolution	1280 x 800 | Recommended: 1920 x 1080
Operating System	Windows 10
Internet Connection	Required for software activation

Installation

Aurora Focus™ can be launched from the Windows 10 Start button.

- Start / All Programs / Zebra Scanner / Zebra Aurora Focus Page

Zebra Aurora Focus™ MV/FIS Configuration Utility

Description

Product Description Zebra Aurora Focus™ is a host-side application which allows for the configuration Zebra FS10, xS20, xS40 and xS70 devices.

Device Compatibility

Compatibility Item	Details
Programming Modes	USB connected cable; M12 Ethernet

Supported Devices	Supported Device FW
FS40, VS40, FS70, VS70	CAAESS00-003-R10
FS10, FS20, VS20	CAAFFS00-003-R10

Release Notes

This document summarizes the following software release:

Software Version Number	Release Date
V6.0.22000	July 2023

V6.0.22000

RELEASE DATE: JULY 2023

- 2D Configuration Barcode Generation and Deployment
- Advanced Filtering and Formatting of Decode Data
- Multi-device management (Up to 5 devices at a time)
    - Firmware Update
    - Backup/Restore
    - Device Log Download
- Zebra Connectivity Gateway for Leader/Followers scanner configuration
- ZETI Command Interface Additional Command Support
- Aurora Focus UX Improvements
    - Improvements to Custom Formatting
    - Improved Localization
- Performance and Enhancements
    - Decode performance improvements.
    - Enhanced Session Based Triggering with Image Save
    - Additional Localization Enhancements

Known Issues

- When saving images over FTP using a hostname, delays in transfer sometimes occur and the image is not transferred.
    - Workaround: Use IP address of FTP server instead.
- When using the Object Plus tools, it will sometimes fail due to the contrast value.
    - Workaround: Adjust contrast value until it succeeds, and the result is satisfactory.
- When a device is in managed mode and backing up from the Aurora Focus Setup Device screen, it will fail.
    - Workaround: Un-mange the device before backing up
- When a device is in managed mode and performing a firmware update up from the Aurora Focus Setup Device screen, it will fail.
    - Workaround: Un-mange the device before doing firmware update
- Rarely OCR performance can be impacted depending on the character set being inspected.
    - Workaround: Adjustment of the size of the ROI can sometimes improve performance
- Some ZETI commands will not work with the command’s shortened format.
    - Workaround: Use the long version of the ZETI command
- When configuring a leader/follower barcode setup, the serial port can sometimes conflict with a job on the device.
    - Workaround: Disable the Serial port in Aurora Focus.
- When deleting cells in the custom formatting tab on VS type devices, sometimes cell will not delete properly, and flickering may occur.
    - Workaround: Move to another tab temporarily and return.

- Flaw detection tools must click ΓÇÿUndo’ twice to perform the undo operation.
  - Workaround: Click undo twice
- When using Aurora Focus for viewing a running barcode job, sometimes decode summary will fail to get updated.
  - Workaround: Close the job being viewed and reopen the job again in view mode.
- In some cases, using many image-banks cause slight delays in processing.
  - Workaround: If possible, reduce number of image banks
- Device will sometimes not be discoverable in Aurora Focus when starting firmware update in HMI until after update is complete and device reboots.
  - Workaround: Wait until firmware update completes.
- Firmware update will fail if memory consumption is already very high.
  - Workaround: Reboot the device before performing firmware update.
- Uninstalling the InstallShield does not always remove all registry entries.
  - Workaround: use `regedit` and manually delete the Aurora Focus folder under `Computer\HKEY_LOCAL_MACHINE\SOFTWARE\Zebra`.

V5.0.13000

RELEASE DATE: FEBRUARY 2023

- Technology Previews
  - Aurora Focus and HMI Language Support - Simplified Chinese, Korean, German, French and Spanish
  - Expanded ZETI Interface Commands
- Tools
  - Deep Learning Based OCR for Fixed Industrial Scanners - Fully Released and Activated via Licenses
- Performance
  - Improvements to deployed job run consistency including Barcode tool timeout, decoding, and trigger timing consistency.
  - Level Trigger Enhancements
  - Triggered Image Buffering
- UX Improvements
  - Backup/Restore/Firmware Upgrade in Aurora Focus Device Configuration 1D Parameter Barcode Setup

V4.0.22000

RELEASE DATE: DECEMBER 2022

- Added Measure Object Tool
- Added Flaw Detection Edge and Intensity Tools
- Added Object Locate Counting
- Added Image Filters: Gradient, Gradient Horizontal, Vertical and Full
- MV Output Formatting on Supported Interfaces (Global or Interface Specific)
- Added Find and Replace to Barcode Decode Tool
- Improved 1D/2D Barcode Decode Performance
- Setting Barcode Match String via TCP or Serial input
- Added Image Download Capability via HMI
- Enhanced Session Based Triggering for Level, Continuous and Presentation Triggers w/Exit Criteria/Aggregation
- Technology Preview: Device Backup/Restore Functionality
- Performance and Stability Enhancements

V3.1.40000

RELEASE DATE: AUGUST 2022

- Added Reset on Trigger for Do Note Read Same Barcode
- Spanish HID Keyboard support
- Performance and Stability Enhancements

V3.0.10000

RELEASE DATE: JUNE 2022

- Added Edge Counting
- Added Deep-Learning Based OCR Tool (*in Beta*)
- Added Circle Find Tool
- Added Circle Diameter Measurement Tool
- Added a Circle ROI Type
- Added Open and Close Filters
- Added Erode Filter Added RS-232 Input Trigger
- Additional 2D Barcode Performance Improvements
- Added ADF to include Pass/Fail Data Types
- Added ADF to include Barcode Quality Metrics Data
- Performance and Stability Enhancements

V2.0.8000

RELEASE DATE: APRIL 2022

- Performance and Stability Enhancements

V2.0.7000

RELEASE DATE: MARCH 2022

- Performance and Stability Enhancements
- 1D Barcode Quality Metrics
- Added Object Locate Plus
- Added Blob Tools
- Added Binarize Filter/Tool
- Added Dilate Filter/Tool
- ImagePerfect(+) fixes & improvements
- Added Scale Calibration for Distance tool
- Added Advanced Pass/Fail Configuration (AND/OR logic for Job results)
- HID Keyboard Expansion - added French and German KB
- Added support for Codabar symbology
- Added Encoding type (Text/Base64) for ADF Full String data type and added option to show nonprintable characters as hex values
- For xS10/20, added external illumination support
- Added "File Path" option when using (S)FTP to Save images
- Added Ability to Open a Local Job (created with emulator) On a Real Connected Camera

V1.200.2001

RELEASE DATE: DECEMBER 2021

- Performance and Stability Enhancements Support for 5MP xS40 and xS70 models
- ImagePerfect(+) for FS models
- Option to output Barcode Location Co-ordinates
- Option to not read last n number of barcodes
- Device can be a TCP/IP Client
- Configurable RS-232 settings
- Ability to configure GPIO Trigger delay
- Support for Fast Hardware Trigger (2.3 MP models only)
- Ability to set GPIOs to static high or low

- Added Job controls to clone, rename and delete jobs
- Support for Conditional Regular expressions for Barcode String Match
- Barcode String Match Regular expression support for ManyCode
- Backwards compatibility of Job files
- Decode and MV statistics in Aurora Focus and HMI stored and queried from device

V1.148.0001

RELEASE DATE: OCTOBER 2021

- Performance Enhancements
- Stability Improvements
- Implementation of licensing error messages in Web HMI
- Fixed issue of Job pausing on refresh of Web HMI
- Fixed issue of crash when clicking on "Communications" tab
- Fixed inability to "View" active Jobs from Job menu when Job utilizes a trigger that is automatic in nature.
- Fixed issue of slow decode rate when "Fast 1D/2D" license is enabled
- Added Configurable TCP/IP - results port and trigger string can be configured
- Support for GPIO > Hardware trigger (normal speed)
- Allow setting a job as Startup / Power up job
- Emulators now have license level
- Supports a total of 21 emulators
- ZJOB backwards compatibility and auto conversion of old job files (Windows only) - Seamless (live) tool changes when Upgrade licenses are applied

V1.134.0001 - Initial Product Release

RELEASE DATE: JULY 2021

- Bug Fixes
- Stability Improvements

Components

SW Component	Version
InstallShield	6.0.22000
Aurora Focus™	6.0.22
BSP (xS40/70)	CAAESS00-003-R10
BSP (FS10/xS20)	CAAFFS00-003-R10
Aurora Focus™ HMI	2.3.70
Connectivity Gateway	1.1.10

The components are installed in the following folders:

Component	Location
Application	Program Files\Zebra Technologies\Zebra Aurora
Job/Configuration Files	Users\Public\Documents\ZebraAurora/ModelType\Jobs
Saved Images	Users\Public\Documents\ZebraAurora/Captures/xSxxxxxx Camera

Minimum PC Requirements

Component	Location
CPU	Intel or AMD 64-bit processor
Random Access Memory Files	4 GB | Recommended: 8GB
GPU	GTX 1030, Intel UHD 620 or equivalent
Graphics Memory	2GB | Recommended: 4GB
Disk Space	4GB free disk space
Monitor Resolution	1280 x 800 | Recommended: 1920 x 1080
Operating System	Windows 10
Internet Connection	Required for software activation

Installation

Aurora Focus™ can be launched from the Windows 10 Start button.

- Start / All Programs / Zebra Scanner / Zebra Aurora Focus

Zebra Aurora™ MV/FIS Configuration Utility

Description

Product Description Zebra Aurora™ is a host-side application which allows for the configuration Zebra FS10, xS20, xS40 and xS70 devices.

Device Compatibility

Compatibility Item	Details
Programming Modes	USB connected cable; M12 Ethernet

Supported Devices	Supported Device FW
FS40, VS40, FS70, VS70	CAAESS00-002-R06
FS10, FS20, VS20	CAAFFS00-002-R06

Release Notes

This document summarizes the following software release:

Software Version Number	Release Date
V5.0.13000	February 2023

V5.0.13000

RELEASE DATE: FEBRUARY 2023

- Technology Previews
  - Aurora and HMI Language Support - Simplified Chinese, Korean, German, French and Spanish
  - Expanded ZETI Interface Commands
- Tools
  - Deep Learning Based OCR for Fixed Industrial Scanners - Fully Released and Activated via Licenses
- Performance
  - Improvements to deployed job run consistency including Barcode tool timeout, decoding, and trigger timing consistency.
  - Level Trigger Enhancements
  - Triggered Image Buffering
- UX Improvements
  - Backup/Restore/Firmware Upgrade in Aurora
- Device Configuration 1D Parameter Barcode Setup

Known Issues

- Rarely, a barcode tool can sometimes fail on certain images if it follows a machine vision tool.
  - Workaround: Move barcode decode tool to run before the machine vision tool.
- When viewing a circle tool that failed in the HMI, it will sometimes cause the HMI to not refresh and will display a blank page.
  - Workaround: Use Aurora to inspect the details of a failing circle tool.
- When a user needs to change a device USB IP address, they must first disable the DHCP server option and then re-enable it again afterwards.
- When executing the ZETI command ΓÇÿTrigger with result’, after using the ΓÇÿDeploy Job’, the trigger command can sometimes fail.
  - Workaround: Add a 2 second delay or more after the ΓÇÿDeploy Job’ command.
- When using ZETI, any command that has a job slot option, the job slot must be specified for this release.

- The ZETI command line ΓÇÿhelp’ command is not complete. Please refer to ZETI documentation for comprehensive overview of the command set
- When in Edit mode in Aurora, setting the GPIO Hardware Trigger falling edge can cause the trigger to use the rising edge instead.
    - Workaround: Set the edge back to rising and then back to falling in the drop-down menu.
- If a device is added to Aurora by IP address, the Aurora Firmware Dual Update will not complete the Dual Update.
    - Workaround: Use HMI if Dual Update is required.
- Flaw detection tools have to click ΓÇÿUndo’ twice to perform the undo operation.
- When using Aurora for viewing a running barcode job, sometimes decode summary will fail to get updated.
    - Workaround: Close the job being viewed and reopen the job again in view mode.
- Device will sometimes not be discoverable in Aurora when starting firmware update in HMI. Device will re-appear once the firmware update completes and reboots
- After updating the firmware, some jobs will not be able to be changed using the HMI.
    - Workaround: Run "User Settings" reset in HMI (must be admin)
- Rarely, the HMI may not appear through the display port unless the device is rebooted.
- Autotune will sometimes fail at a range of ~200mm on 2D barcodes only, if illumination is off
    - Workaround: Ensure adequate light is available, either with continuous light or enabling the LED lights on the device.
- When generating a backup, if done after setting the time zone, the backup may fail.
    - Workaround: Reboot device after time zone change if planning to backup.
- When setting the TCP Heartbeat string in Aurora, nonstandard ASCI characters may cause the heartbeat to not be sent.
    - Workaround: Ensure any heartbeat string the user defines uses standard ASCII characters only
- When reordering tools in Aurora that have dependencies on other tools, the reordered tool may not be able to be edited.
    - Workaround: Remove and re-add the tool back to the workflow.
- Certain special characters in a job name cause the job results image folder to not allow download from the device.
    - Workaround: Use only standard characters for job names
- In some cases, using large number of image banks cause slight delays in processing.
- After upgrading the device firmware, USB options may be grayed out.
    - Workaround: Perform User Setting Reset from HMI and update device settings after.
- Firmware update will fail if memory consumption is already high.
    - Workaround: Reboot the device and upgrade
- Uninstalling the InstallShield does not always remove all registry entries.
    - Workaround: use `regedit` and manually delete Aurora folder under `Computer\HKEY_LOCAL_MACHINE\SOFTWARE\Zebra`.

V4.0.22000

RELEASE DATE: DECEMBER 2022

- Added Measure Object Tool
- Added Flaw Detection Edge and Intensity Tools
- Added Object Locate Counting
- Added Image Filters: Gradient, Gradient Horizontal, Vertical and Full
- MV Output Formatting on Supported Interfaces (Global or Interface Specific)
- Added Find and Replace to Barcode Decode Tool
- Improved 1D/2D Barcode Decode Performance
- Setting Barcode Match String via TCP or Serial input
- Added Image Download Capability via HMI
- Enhanced Session Based Triggering for Level, Continuous and Presentation Triggers w/Exit Criteria/Aggregation
- Technology Preview: Device Backup/Restore Functionality
- Performance and Stability Enhancements

V3.1.40000

RELEASE DATE: AUGUST 2022

- Added Reset on Trigger for Do Note Read Same Barcode
- Spanish HID Keyboard support
- Performance and Stability Enhancements

V3.0.10000

RELEASE DATE: JUNE 2022

- Added Edge Counting
- Added Deep-Learning Based OCR Tool (*in Beta*)
- Added Circle Find Tool
- Added Circle Diameter Measurement Tool
- Added a Circle ROI Type
- Added Open and Close Filters
- Added Erode Filter
- Added RS-232 Input Trigger
- Additional 2D Barcode Performance Improvements
- Added ADF to include Pass/Fail Data Types
- Added ADF to include Barcode Quality Metrics Data
- Performance and Stability Enhancements

V2.0.8000

RELEASE DATE: APRIL 2022

- Performance and Stability Enhancements

V2.0.7000

RELEASE DATE: MARCH 2022

- Performance and Stability Enhancements
- 1D Barcode Quality Metrics
- Added Object Locate Plus
- Added Blob Tools
- Added Binarize Filter/Tool
- Added Dilate Filter/Tool
- ImagePerfect(+) fixes & improvements
- Added Scale Calibration for Distance tool
- Added Advanced Pass/Fail Configuration (AND/OR logic for Job results)
- HID Keyboard Expansion - added French and German KB
- Added support for Codabar symbology
- Added Encoding type (Text/Base64) for ADF Full String data type and added option to show nonprintable characters as hex values
- For xS10/20, added external illumination support
- Added "File Path" option when using (S)FTP to Save images
- Added Ability to Open a Local Job (created with emulator) On a Real Connected Camera

V1.200.2001

RELEASE DATE: DECEMBER 2021

- Performance and Stability Enhancements
- Support for 5MP xS40 and xS70 models
- ImagePerfect(+) for FS models
- Option to output Barcode Location Co-ordinates
- Option to not read last n number of barcodes
- Device can be a TCP/IP Client
- Configurable RS-232 settings
- Ability to configure GPIO Trigger delay

- Support for Fast Hardware Trigger (2.3 MP models only)
- Ability to set GPIOs to static high or low
- Added Job controls to clone, rename and delete jobs
- Support for Conditional Regular expressions for Barcode String Match
- Barcode String Match Regular expression support for ManyCode
- Backwards compatibility of Job files
- Decode and MV statistics in Aurora and HMI stored and queried from device

V1.148.0001

RELEASE DATE: OCTOBER 2021

- Performance Enhancements
- Stability Improvements
- Implementation of licensing error messages in Web HMI
- Fixed issue of Job pausing on refresh of Web HMI
- Fixed issue of crash when clicking on "Communications" tab
- Fixed inability to "View" active Jobs from Job menu when Job utilizes a trigger that is automatic in nature.
- Fixed issue of slow decode rate when "Fast 1D/2D" license is enabled
- Added Configurable TCP/IP - results port and trigger string can be configured
- Support for GPIO > Hardware trigger (normal speed)
- Allow setting a job as Startup / Power up job
- Emulators now have license level
- Supports a total of 21 emulators
- ZJOB backwards compatibility and auto conversion of old job files (Windows only) - Seamless (live) tool changes when Upgrade licenses are applied

V1.134.0001 - Initial Product Release

RELEASE DATE: JULY 2021

- Bug Fixes
- Stability Improvements

Components

SW Component	Version
InstallShield	5.0.13000
Aurora™	5.0.13
BSP (xS40/70)	CAAESS00-002-R06
BSP (FS10/xS20)	CAAFFS00-002-R06
Aurora™ HMI	2.1.00

The components are installed in the following folders:

Component	Location
Application	Program Files\Zebra Technologies\Aurora
Job/Configuration Files	Users\Public\Documents\ZebraAurora\xSx0\Jobs
Saved Images	Users\Public\Documents\ZebraAurora\Captures\xSxxxxxx Camera

Minimum PC Requirements

Component	Location
CPU	Intel or AMD 64-bit processor
Random Access Memory Files	4 GB | Recommended: 8GB
GPU	GTX 1030, Intel UHD 620 or equivalent
Graphics Memory	2GB | Recommended: 4GB
Disk Space	4GB free disk space
Monitor Resolution	1280 x 800 | Recommended: 1920 x 1080
Operating System	Windows 10
Internet Connection	Required for software activation

Installation

Aurora™ can be launched from the Windows 10 Start button.

- Start / All Programs / Zebra Scanner / Zebra Aurora

Zebra Aurora™ MV/FIS Configuration Utility

Description

Product Description Zebra Aurora™ is a host-side application which allows for the configuration Zebra FS10, xS20, xS40 and xS70 devices.

Device Compatibility

Compatibility Item	Details
Programming Modes	USB connected cable; M12 Ethernet

Supported Devices	Supported Device FW
FS40, VS40, FS70, VS70	CAAESS00-001-R53
FS10, FS20, VS20	CAAFFS00-001-R53

Release Notes

This document summarizes the following software release:

Software Version Number	Release Date
V4.0.23000	December 2022

V4.0.23000

RELEASE DATE: DECEMBER 2022

- Performance and Stability Enhancements

V4.0.22000

RELEASE DATE: DECEMBER 2022

- Added Measure Object Tool
- Added Flaw Detection Edge and Intensity Tools
- Added Object Locate Counting
- Added Image Filters: Gradient, Gradient Horizontal, Vertical and Full
- MV Output Formatting on Supported Interfaces (Global or Interface Specific)
- Added Find and Replace to Barcode Decode Tool
- Improved 1D/2D Barcode Decode Performance
- Setting Barcode Match String via TCP or Serial input
- Added Image Download Capability via HMI
- Enhanced Session Based Triggering for Level, Continuous and Presentation Triggers w/Exit Criteria/Aggregation
- Technology Preview: Device Backup/Restore Functionality
- Performance and Stability Enhancements

V3.1.40000

RELEASE DATE: AUGUST 2022

- Added Reset on Trigger for Do Note Read Same Barcode

- Spanish HID Keyboard support
- Performance and Stability Enhancements

V3.0.10000

RELEASE DATE: JUNE 2022

- Added Edge Counting
- Added Deep-Learning Based OCR Tool (*in Beta*)
- Added Circle Find Tool
- Added Circle Diameter Measurement Tool
- Added a Circle ROI Type
- Added Open and Close Filters
- Added Erode Filter
- Added RS-232 Input Trigger
- Additional 2D Barcode Performance Improvements
- Added ADF to include Pass/Fail Data Types
- Added ADF to include Barcode Quality Metrics Data
- Performance and Stability Enhancements

V2.0.8000

RELEASE DATE: APRIL 2022

- Performance and Stability Enhancements

V2.0.7000

RELEASE DATE: MARCH 2022

- Performance and Stability Enhancements
- 1D Barcode Quality Metrics
- Added Object Locate Plus
- Added Blob Tools
- Added Binarize Filter/Tool
- Added Dilate Filter/Tool
- ImagePerfect(+) fixes & improvements
- Added Scale Calibration for Distance tool
- Added Advanced Pass/Fail Configuration (AND/OR logic for Job results)
- HID Keyboard Expansion - added French and German KB
- Added support for Codabar symbology
- Added Encoding type (Text/Base64) for ADF Full String data type and added option to show nonprintable characters as hex values
- For xS10/20, added external illumination support
- Added "File Path" option when using (S)FTP to Save images
- Added Ability to Open a Local Job (created with emulator) On a Real Connected Camera

V1.200.2001

RELEASE DATE: DECEMBER 2021

- Performance and Stability Enhancements
- Support for 5MP xS40 and xS70 models
- ImagePerfect(+) for FS models
- Option to output Barcode Location Co-ordinates
- Option to not read last n number of barcodes

- Device can be a TCP/IP Client
- Configurable RS-232 settings
- Ability to configure GPIO Trigger delay
- Support for Fast Hardware Trigger (2.3 MP models only)
- Ability to set GPIOs to static high or low
- Added Job controls to clone, rename and delete jobs
- Support for Conditional Regular expressions for Barcode String Match
- Barcode String Match Regular expression support for ManyCode
- Backwards compatibility of Job files
- Decode and MV statistics in Aurora and HMI stored and queried from device

V1.148.0001

RELEASE DATE: OCTOBER 2021

- Performance Enhancements
- Stability Improvements
- Implementation of licensing error messages in Web HMI
- Fixed issue of Job pausing on refresh of Web HMI
- Fixed issue of crash when clicking on "Communications" tab
- Fixed inability to "View" active Jobs from Job menu when Job utilizes a trigger that is automatic in nature.
- Fixed issue of slow decode rate when "Fast 1D/2D" license is enabled
- Added Configurable TCP/IP - results port and trigger string can be configured
- Support for GPIO > Hardware trigger (normal speed)
- Allow setting a job as Startup / Power up job
- Emulators now have license level
- Supports a total of 21 emulators
- ZJOB backwards compatibility and auto conversion of old job files (Windows only) - Seamless (live) tool changes when Upgrade licenses are applied

V1.134.0001 - Initial Product Release

RELEASE DATE: JULY 2021

- Bug Fixes
- Stability Improvements

Known Issues

- Backup and restore requires the device hostname to start with proper the model type (VS or FS)
- Certain special characters in a job name cause the job results image folder to not allow download from the device.
    - Workaround: Use only standard characters for job names
- Locate Edge timeout value is sometimes not honored when timeout value is very low and searching for low contrast edges and/or high number of edges.
- In some cases, using large number of image banks cause slight delays in processing.
- When changing USB options during setup: Enabling/Disabling HID or changing CDC serial options, the USB cable needs to be plugged in; otherwise, the device will need to be restarted after the changes are made.
- Changing the scale or size of an item in the flaw detection tool, when used in conjunction with fixturing with Locate Object Plus, requires readjustment of the ROI when retraining to the new size.
    - Workaround: Adjust ROI and Retrain or use Standard (non-Plus) Locate Object tool.
- After upgrading the device firmware, USB options may be grayed out.
    - Workaround: Perform User Setting Reset from HMI and update device settings after.
- Firmware update will fail if memory consumption is already high.
    - Workaround: Reboot the device and upgrade
- Uninstalling the InstallShield does not always remove all registry entries.
    - Workaround: regedit and manually delete Aurora folder under Computer\HKEY_LOCAL_MACHINE\SOFTWARE\Zebra

Components

SW Component	Version
InstallShield	4.0.23000
Aurora™	4.0.23
BSP (xS40/70)	CAAESS00-001-R53
BSP (FS10/xS20)	CAAFFS00-001-R53
Aurora™ HMI	2.0.00

The components are installed in the following folders:

Component	Location
Application	Program Files\Zebra Technologies\Aurora
Job/Configuration Files	Users\Public\Documents\ZebraAurora\xSx0\Jobs
Saved Images	Users\Public\Documents\ZebraAurora\Captures\xSxxxxxx Camera

Minimum PC Requirements

Component	Location
CPU	Intel or AMD 64-bit processor
Random Access Memory Files	4 GB | Recommended: 8GB
GPU	GTX 1030, Intel UHD 620 or equivalent
Graphics Memory	2GB | Recommended: 4GB
Disk Space	4GB free disk space
Monitor Resolution	1280 x 800 | Recommended: 1920 x 1080
Operating System	Windows 10
Internet Connection	Required for software activation

Installation

Aurora™ can be launched from the Windows 10 Start button.

- Start / All Programs / Zebra Scanner / Zebra Aurora

Zebra Aurora™ MV/FIS Configuration Utility

Description

Product Description Zebra Aurora™ is a host-side application which allows for the configuration Zebra FS10, xS20, xS40 and xS70 devices.

Device Compatibility

Compatibility Item	Details
Programming Modes	USB connected cable; M12 Ethernet

Supported Devices	Supported Device FW
FS40, VS40, FS70, VS70	CAAESS00-001-R45
FS10, FS20, VS20	CAAFFS00-001-R45

Release Notes

This document summarizes the following software release:

Software Version Number	Release Date
V3.1.40000	August 2022

V3.1.40000

RELEASE DATE: AUGUST 2022

- Added Reset on Trigger for Do Note Read Same Barcode
- Spanish HID Keyboard support
- Performance and Stability Enhancements

V3.0.10000

RELEASE DATE: JUNE 2022

- Added Edge Counting
- Added Deep-Learning Based OCR Tool (*in Beta*)
- Added Circle Find Tool
- Added Circle Diameter Measurement Tool
- Added a Circle ROI Type
- Added Open and Close Filters
- Added Erode Filter
- Added RS-232 Input Trigger
- Additional 2D Barcode Performance Improvements
- Added ADF to include Pass/Fail Data Types
- Added ADF to include Barcode Quality Metrics Data
- Performance and Stability Enhancements

V2.0.8000

RELEASE DATE: APRIL 2022

- Performance and Stability Enhancements

V2.0.7000

RELEASE DATE: MARCH 2022

- Performance and Stability Enhancements
- 1D Barcode Quality Metrics
- Added Object Locate Plus
- Added Blob Tools
- Added Binarize Filter/Tool
- Added Dilate Filter/Tool
- ImagePerfect(+) fixes & improvements
- Added Scale Calibration for Distance tool
- Added Advanced Pass/Fail Configuration (AND/OR logic for Job results)
- HID Keyboard Expansion - added French and German KB
- Added support for Codabar symbology
- Added Encoding type (Text/Base64) for ADF Full String data type and added option to show nonprintable characters as hex values
- For xS10/20, added external illumination support
- Added "File Path" option when using (S)FTP to Save images
- Added Ability to Open a Local Job (created with emulator) On a Real Connected Camera

V1.200.2001

RELEASE DATE: DECEMBER 2021

- Performance and Stability Enhancements
- Support for 5MP xS40 and xS70 models
- ImagePerfect(+) for FS models
- Option to output Barcode Location Co-ordinates
- Option to not read last n number of barcodes
- Device can be a TCP/IP Client
- Configurable RS-232 settings
- Ability to configure GPIO Trigger delay

- Support for Fast Hardware Trigger (2.3 MP models only)
- Ability to set GPIOs to static high or low
- Added Job controls to clone, rename and delete jobs
- Support for Conditional Regular expressions for Barcode String Match
- Barcode String Match Regular expression support for ManyCode
- Backwards compatibility of Job files
- Decode and MV statistics in Aurora and HMI stored and queried from device

V1.148.0001

RELEASE DATE: OCTOBER 2021

- Performance Enhancements
- Stability Improvements
- Implementation of licensing error messages in Web HMI
- Fixed issue of Job pausing on refresh of Web HMI
- Fixed issue of crash when clicking on "Communications" tab
- Fixed inability to "View" active Jobs from Job menu when Job utilizes a trigger that is automatic in nature.
- Fixed issue of slow decode rate when "Fast 1D/2D" license is enabled
- Added Configurable TCP/IP - results port and trigger string can be configured
- Support for GPIO > Hardware trigger (normal speed)
- Allow setting a job as Startup / Power up job
- Emulators now have license level
- Supports a total of 21 emulators
- ZJOB backwards compatibility and auto conversion of old job files (Windows only) - Seamless (live) tool changes when Upgrade licenses are applied

V1.134.0001 - Initial Product Release

RELEASE DATE: JULY 2021

- Bug Fixes
- Stability Improvements

Known Issues

- When editing a job on the camera, it can sometimes take up to 30 seconds for the Edit mode to complete, if the job has been running for a long time.
- When using Profinet and running a job for a long time, users should not use the HMI Dashboard View and instead use the Live Monitoring View. The Dashboard View can sometimes impact job performance.
- When using custom trigger string, using the NULL "\0" terminator is not working.
- The Data code tool does not yet support Decode time on results for TCP and other Interfaces
- When modifying certain parameters on Locate Object tools, when object is complex or large, an additional delay occurs. This occurs during setup only.
- When fixturing to the Circle Tool, if the found circle changes from the original expected size, this can cause a shift the location of the fixtured ROI.
- Switching between different HID modes (USB/CDC and HID-KB) can sometimes take extended time.
- External illumination does not turn ON when running a job created with a previous (noncurrent) firmware version.
- Workaround: Enable/Reenable External Illumination
- In some cases, External Illumination cannot be turned off when in NPN mode
    - Workaround: Switch back to PNP mode before turning off
- On FS10, certain hubs cause USB interfaces (both RNDIS and HID) to break on HID Keyboard re-enumeration
    - Workaround: Disconnect and reconnect USB cable.
- Firmware update is failing with "Invalid plugin file" when FW plugin filename has special characters (i.e. space , parentheses, etc)
- Firmware update will fail if memory consumption is already high.
- Workaround: Reboot the device and upgrade
- "Image Saving" logic does not take into account "Advanced Pass/Fail Configuration" or local "Disabling" of tool
- Uninstalling the InstallShield does not always remove all registry entries.

- Workaround: regedit and manually delete Aurora folder under Computer\HKEY_LOCAL_MACHINE\SOFTWARE\Zebra\

Components

SW Component	Version
InstallShield	3.1.40000
Aurora™	3.1.40
BSP (xS40/70)	CAAESS00-001-R45
BSP (FS10/xS20)	CAAFFS00-001-R45
Aurora™ HMI	1.0.44

The components are installed in the following folders:

Component	Location
Application	Program Files\Zebra Technologies\Aurora
Job/Configuration Files	Users\Public\Documents\ZebraAurora\xSx0\Jobs
Saved Images	Users\Public\Documents\ZebraAurora\Captures\xSxxxxxx Camera

Minimum PC Requirements

Component	Location
CPU	Intel or AMD 64-bit processor
Random Access Memory Files	4 GB | Recommended: 8GB
GPU	GTX 1030, Intel UHD 620 or equivalent
Graphics Memory	2GB | Recommended: 4GB
Disk Space	4GB free disk space
Monitor Resolution	1280 x 800 | Recommended: 1920 x 1080
Operating System	Windows 10
Internet Connection	Required for software activation

Installation

Aurora™ can be launched from the Windows 10 Start button.

- Start / All Programs / Zebra Scanner / Zebra Aurora

Zebra Aurora™ MV/FIS Configuration Utility

Description

Product Description Zebra Aurora™ is a host-side application which allows for the configuration Zebra FS10, xS20, xS40 and xS70 devices.

Device Compatibility

Compatibility Item	Details
Programming Modes	USB connected cable; M12 Ethernet

Supported Devices	Supported Device FW
FS40, VS40, FS70, VS70	CAAESS00-001-R38
FS10, FS20, VS20	CAAFFS00-001-R38

Release Notes

This document summarizes the following software release:

Software Version Number	Release Date
V3.0.10000	June 2022

V3.0.10000

RELEASE DATE: JUNE 2022

- Added Edge Counting
- Added Deep-Learning Based OCR Tool (*in Beta*)
- Added Circle Find Tool
- Added Circle Diameter Measurement Tool
- Added a Circle ROI Type
- Added Open and Close Filters
- Added Erode Filter
- Added RS-232 Input Trigger
- Additional 2D Barcode Performance Improvements
- Added ADF to include Pass/Fail Data Types

- Added ADF to include Barcode Quality Metrics Data
- Performance and Stability Enhancements

V2.0.8000

RELEASE DATE: APRIL 2022

- Performance and Stability Enhancements

V2.0.7000

RELEASE DATE: MARCH 2022

- Performance and Stability Enhancements
- 1D Barcode Quality Metrics
- Added Object Locate Plus
- Added Blob Tools
- Added Binarize Filter/Tool
- Added Dilate Filter/Tool
- ImagePerfect(+) fixes & improvements
- Added Scale Calibration for Distance tool
- Added Advanced Pass/Fail Configuration (AND/OR logic for Job results)
- HID Keyboard Expansion - added French and German KB
- Added support for Codabar symbology
- Added Encoding type (Text/Base64) for ADF Full String data type and added option to show nonprintable characters as hex values
- For xS10/20, added external illumination support
- Added "File Path" option when using (S)FTP to Save images
- Added Ability to Open a Local Job (created with emulator) On a Real Connected Camera

V1.200.2001

RELEASE DATE: DECEMBER 2021

- Performance and Stability Enhancements
- Support for 5MP xS40 and xS70 models
- ImagePerfect(+) for FS models
- Option to output Barcode Location Co-ordinates
- Option to not read last n number of barcodes
- Device can be a TCP/IP Client
- Configurable RS-232 settings
- Ability to configure GPIO Trigger delay
- Support for Fast Hardware Trigger (2.3 MP models only)
- Ability to set GPIOs to static high or low
- Added Job controls to clone, rename and delete jobs
- Support for Conditional Regular expressions for Barcode String Match
- Barcode String Match Regular expression support for ManyCode
- Backwards compatibility of Job files
- Decode and MV statistics in Aurora and HMI stored and queried from device

V1.148.0001

RELEASE DATE: OCTOBER 2021

- Performance Enhancements
- Stability Improvements

- Implementation of licensing error messages in Web HMI
- Fixed issue of Job pausing on refresh of Web HMI
- Fixed issue of crash when clicking on "Communications" tab
- Fixed inability to "View" active Jobs from Job menu when Job utilizes a trigger that is automatic in nature.
- Fixed issue of slow decode rate when "Fast 1D/2D" license is enabled
- Added Configurable TCP/IP - results port and trigger string can be configured
- Support for GPIO > Hardware trigger (normal speed)
- Allow setting a job as Startup / Power up job
- Emulators now have license level
- Supports a total of 21 emulators
- ZJOB backwards compatibility and auto conversion of old job files (Windows only)
- Seamless (live) tool changes when Upgrade licenses are applied

V1.134.0001 - Initial Product Release

RELEASE DATE: JULY 2021

- Bug Fixes
- Stability Improvements

Known Issues

- When using fixturing against a Circle Tool, during setup, the fixtured tool will sometimes move and will have to be manually adjusted before deploying the job.
- Switching between different HID modes (USB/CDC and HID-KB) can sometimes take extended time.
- External illumination does not turn ON when running a job created with a previous (noncurrent) firmware version.
- Workaround: Enable/Reenable External Illumination
- In some cases, External Illumination cannot be turned off when in NPN mode
    - Workaround: Switch back to PNP mode before turning off
- On FS10, certain hubs cause USB interfaces (both RNDIS and HID) to break on HID Keyboard re-enumeration
    - Workaround: Disconnect and reconnect USB cable.
- Firmware update is failing with "Invalid plugin file" when FW plugin filename has special characters (i.e. space , parentheses, etc)
- Firmware update will fail if memory consumption is already high.
- Workaround: Reboot the device and upgrade
- "Image Saving" logic does not take into account "Advanced Pass/Fail Configuration" or local "Disabling" of tool
- Uninstalling the InstallShield does not always remove all registry entries.
- Workaround: regedit and manually delete Aurora folder under Computer\HKEY_LOCAL_MACHINE\SOFTWARE\Zebra\

Components

SW Component	Version
InstallShield	3.0.10000
Aurora™	3.0.10
BSP (xS40/70)	CAAESS00-001-R38
BSP (FS10/xS20)	CAAFFS00-001-R38
Aurora™ HMI	1.0.43

The components are installed in the following folders:

Component	Location
Application	Program Files\Zebra Technologies\Aurora
Job/Configuration Files	Users\Public\Documents\ZebraAurora\xSx0\Jobs
Saved Images	Users\Public\Documents\ZebraAurora\Captures\xSxxxxxx Camera

Minimum PC Requirements

Component	Location
CPU	Intel or AMD 64-bit processor
Random Access Memory Files	4 GB | Recommended: 8GB
GPU	GTX 1030, Intel UHD 620 or equivalent
Graphics Memory	2GB | Recommended: 4GB
Disk Space	4GB free disk space
Monitor Resolution	1280 x 800 | Recommended: 1920 x 1080
Operating System	Windows 10
Internet Connection	Required for software activation

Installation

Aurora™ can be launched from the Windows 10 Start button.

- Start / All Programs / Zebra Scanner / Zebra Aurora

Zebra Aurora™ MV/FIS Configuration Utility

Description

Product Description Zebra Aurora™ is a host-side application which allows for the configuration Zebra FS10, xS20, xS40 and xS70 devices.

Device Compatibility

Compatibility Item	Details
Programming Modes	USB connected cable; M12 Ethernet

Supported Devices	Supported Device FW
FS40, VS40, FS70, VS70	CAAESS00-001-R34
FS10, FS20, VS20	CAAFFS00-001-R34

Release Notes

This document summarizes the following software release:

Software Version Number	Release Date
V2.0.8000	April 2022

V2.0.8000

RELEASE DATE: APRIL 2022

- Performance and Stability Enhancements

V2.0.7000

RELEASE DATE: MARCH 2022

- Performance and Stability Enhancements
- 1D Barcode Quality Metrics
- Added Object Locate Plus
- Added Blob Tools
- Added Binarize Filter/Tool
- Added Dilate Filter/Tool
- ImagePerfect(+) fixes & improvements
- Added Scale Calibration for Distance tool
- Added Advanced Pass/Fail Configuration (AND/OR logic for Job results)
- HID Keyboard Expansion - added French and German KB
- Added support for Codabar symbology
- Added Encoding type (Text/Base64) for ADF Full String data type and added option to show nonprintable characters as hex values
- For xS10/20, added external illumination support
- Added "File Path" option when using (S)FTP to Save images
- Added Ability to Open a Local Job (created with emulator) On a Real Connected Camera

V1.200.2001

RELEASE DATE: DECEMBER 2021

- Performance and Stability Enhancements
- Support for 5MP xS40 and xS70 models
- ImagePerfect(+) for FS models
- Option to output Barcode Location Co-ordinates
- Option to not read last n number of barcodes
- Device can be a TCP/IP Client
- Configurable RS-232 settings
- Ability to configure GPIO Trigger delay
- Support for Fast Hardware Trigger (2.3 MP models only)
- Ability to set GPIOs to static high or low
- Added Job controls to clone, rename and delete jobs
- Support for Conditional Regular expressions for Barcode String Match
- Barcode String Match Regular expression support for ManyCode
- Backwards compatibility of Job files
- Decode and MV statistics in Aurora and HMI stored and queried from device

V1.148.0001

RELEASE DATE: OCTOBER 2021

- Performance Enhancements
- Stability Improvements
- Implementation of licensing error messages in Web HMI
- Fixed issue of Job pausing on refresh of Web HMI
- Fixed issue of crash when clicking on "Communications" tab
- Fixed inability to "View" active Jobs from Job menu when Job utilizes a trigger that is automatic in nature.
- Fixed issue of slow decode rate when "Fast 1D/2D" license is enabled
- Added Configurable TCP/IP - results port and trigger string can be configured
- Support for GPIO > Hardware trigger (normal speed)
- Allow setting a job as Startup / Power up job
- Emulators now have license level
- Supports a total of 21 emulators
- ZJOB backwards compatibility and auto conversion of old job files (Windows only) - Seamless (live) tool changes when Upgrade licenses are applied

V1.134.0001 - Initial Product Release

RELEASE DATE: JULY 2021

- Bug Fixes
- Stability Improvements

Known Issues

- External illumination does not turn ON when running a job created with a previous (noncurrent) firmware version
- Continuous and Presentation modes are disabled for VS devices when only "Read DPM" or "Read DPM & Barcode" tool is added in flow builder
- Mouse can stop responding on DisplayPort after prolonged use.
- Workaround: press "Ctrl + Alt + Backspace" to restart HMI
- On FS10, certain hubs cause USB interfaces (both RNDIS and HID) to break on HID Keyboard re-enumeration
    - Workaround: Disconnect and reconnect USB cable.
- Within "Locate Object Plus" tool, the edge contrast image fails to update when tool fails

- Firmware update is failing with "Invalid plugin file" when FW plugin filename has special characters (i.e. space , parentheses, etc)
- Firmware update will fail if memory consumption is already high
- "Image Saving" logic does not take into account "Advanced Pass/Fail Configuration" or local "Disabling" of tool
- Startup job can change when FW Update is performed when a Continuous job is running
- Monitoring device via Aurora for a long time may result in an Aurora restart.
    - Recommended: Use Web HMI for long term job monitoring
- Uninstalling the InstallShield does not always remove all registry entries.
- Workaround: regedit and manually delete Aurora folder under Computer\HKEY_LOCAL_MACHINE\SOFTWARE\Zebra\
- FS40 jobs may disconnect and fall back to emulator-mode when "Show Images" option is disabled and "Save Images" option is enabled while powered via 24v power supply and using USB-C communication cable.

Components

SW Component	Version
InstallShield	2.0.8000
Aurora™	2.0.8
BSP (xS40/70)	CAAESS00-001-R34
BSP (FS10/xS20)	CAAFFS00-001-R34
Aurora™ HMI	1.0.35

The components are installed in the following folders:

Component	Location
Application	Program Files\Zebra Technologies\Aurora
Job/Configuration Files	Users\Public\Documents\ZebraAurora\xSx0\Jobs
Saved Images	Users\Public\Documents\ZebraAurora\Captures\xSxxxxxx Camera

Minimum PC Requirements

Component	Location
CPU	Intel or AMD 64-bit processor
Random Access Memory Files	4 GB | Recommended: 8GB
GPU	GTX 1030, Intel UHD 620 or equivalent
Graphics Memory	2GB | Recommended: 4GB
Disk Space	4GB free disk space
Monitor Resolution	1280 x 800 | Recommended: 1920 x 1080
Operating System	Windows 10
Internet Connection	Required for software activation

Installation

Aurora™ can be launched from the Windows 10 Start button.

- Start / All Programs / Zebra Scanner / Zebra Aurora

Zebra Aurora™ MV/FIS Configuration Utility

Description

Product Description Zebra Aurora™ is a host-side application which allows for the configuration Zebra FS10, xS20, xS40 and xS70 devices.

Device Compatibility

Compatibility Item	Details
Programming Modes	USB connected cable; TCP/IP (Ethernet)

Supported Devices	Supported Device FW
FS10, FS20, VS20, FS40, VS40, FS70, VS70	CAAESS00-001-R28

Release Notes

This document summarizes the following software release:

Software Version Number	Release Date
V1.200.2001	December 2021

V1.200.2001

RELEASE DATE: DECEMBER 2021

- Performance and Stability Enhancements
- Support for 5MP xS40 and xS70 models
- ImagePerfect(+) for FS models
- Option to output Barcode Location Co-ordinates
- Option to not read last n number of barcodes
- Device can be a TCP/IP Client
- Configurable RS-232 settings
- Ability to configure GPIO Trigger delay
- Support for Fast Hardware Trigger (2.3 MP models only)
- Ability to set GPIOs to static high or low
- Added Job controls to clone, rename and delete jobs
- Support for Conditional Regular expressions for Barcode String Match
- Barcode String Match Regular expression support for ManyCode
- Backwards compatibility of Job files
- Decode and MV statistics in Aurora and HMI stored and queried from device

V1.148.0001

RELEASE DATE: OCTOBER 2021

- Performance Enhancements
- Stability Improvements
- Implementation of licensing error messages in Web HMI
- Fixed issue of Job pausing on refresh of Web HMI
- Fixed issue of crash when clicking on "Communications" tab
- Fixed inability to "View" active Jobs from Job menu when Job utilizes a trigger that is automatic in nature.
- Fixed issue of slow decode rate when "Fast 1D/2D" license is enabled
- Added Configurable TCP/IP - results port and trigger string can be configured
- Support for GPIO > Hardware trigger (normal speed)
- Allow setting a job as Startup / Power up job
- Emulators now have license level
- Supports a total of 21 emulators
- ZJOB backwards compatibility and auto conversion of old job files (Windows only) - Seamless (live) tool changes when Upgrade licenses are applied

V1.134.0001 - Initial Product Release

RELEASE DATE: JULY 2021

- Bug Fixes
- Stability Improvements

Known Issues

- Factory reset required for FS10/xS20 after updating to this release
- Numeric entry fields can sometimes miss keystrokes when typing or overwrite characters
- Able to deploy a Locate Object job while the tool is in the training state
- Presentation Mode increments No-Read counter on successful decode
- Deploy to Edit mode transition during Pulse Width of a GPIO output causes the GPIO to stay ON
- External light can stay in Always On Mode when a certain order of operations is performed
- Mouse can stop responding on DisplayPort after prolonged use. Workaround: press "Ctrl + Alt + Backspace" to restart HMI
- WebHmi: Columns in Results History page changes on refresh of the page and can result in a white screen on clicking View Results button
- Startup job can change when FW Update is performed when a Continuous job is running
- Fixturing to Locate Edge tool works incorrectly if the target moves before the first trigger. Workaround: Do not move the target for the first trigger after Deploy
- Running a Continuous job with 0ms Same Data Timeout and Show Images can cause the job to stop after several days of running. Workaround: Uncheck Show Image in Capture > Triggers page
- Monitoring device via Aurora for a long time may result in an Aurora restart. Recommended: Use Web HMI for long term job monitoring
- ImagePerfect(+) for FS models does not work the exact same way in Edit mode as it does in Deploy mode in certain ordering of Banks
- Uninstalling the InstallShield does not always remove all registry entries. Workaround: regedit and manually delete folders 123MV and Aurora under Computer\HKEY_LOCAL_MACHINE\SOFTWARE\Zebra\
- Certain unsupported settings (RS232, GPIO) are displayed in Aurora for FS10 model
- Certain hubs cause USB interfaces (both RNDIS and HID) on FS10 to break on HID Keyboard re-enumeration. Workaround: power cycle device
- When Long Exposure is checked in FS10/xS20 illumination flashes twice for every capture and trigger
- Local Adapter drop down in Connect > Save Image > (S)FTP will be deprecated in a future release, since incorrect configuration can lead to device disconnections from Aurora.
- Image Capture on FS10/xS20 can take a few seconds to complete occasionally (in particular on first time connection to device)
- Presentation Mode for FS10/xS20 is a Beta feature
- If Barcode tool's search ROI is mostly off the image it may cause Aurora to restart

Components

SW Component	Version
InstallShield	1.200.2001
Aurora™	1.2.20
BSP (xS40/70)	CAAESS00-001-R28
BSP (FS10/xS20)	CAAFFS00-001-R03
Aurora™ HMI	1.0.31

The components are installed in the following folders:

Component	Location
Application	Program Files\Zebra Technologies\Aurora
Job/Configuration Files	Users\Public\Documents\ZebraAurora\xSx0\Jobs
Saved Images	Users\Public\Documents\ZebraAurora\Captures\xSxxxxxx Camera

Minimum PC Requirements

Component	Location
CPU	Intel or AMD 64-bit processor
Random Access Memory Files	4 GB | Recommended: 8GB
GPU	GTX 1030, Intel UHD 620 or equivalent
Graphics Memory	2GB | Recommended: 4GB
Disk Space	4GB free disk space
Monitor Resolution	1280 x 800 | Recommended: 1920 x 1080
Operating System	Windows 10
Internet Connection	Required for software activation

Installation

Aurora™ can be launched from the Windows 10 Start button.

- Start / All Programs / Zebra Scanner / Zebra Aurora